A Measure-Theoretic Analysis of Reasoning in Transformers: Structural Generalization and Approximation Limits

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Abstract

Scaling laws for reasoning in large language models (LLMs) are well documented, but little theory explains their out-of-distribution (OOD) generalization. We formalize reasoning with optimal transport: each search trajectory is mapped to twelve structural coordinates, such as tree depth and backtracking rate, and dataset shift is measured by the Wasserstein-1 distance. Kantorovich duality bounds the OOD risk through this distance, the architecture's Lipschitz constant and approximation limits. First, under absolute positional encoding (APE), shifting part of a trajectory to later positions almost surely changes the attention scores within that part; if this misroutes exact-match search, the Lipschitz constant and risk are $\Omega(1)$. Shift-invariant encodings such as rotary position embedding (RoPE) leave the scores unchanged and bound the error. Second, we map backtracking to a Dyck-k language and, under an implicit-depth assumption, derive a risk floor for search trees deeper than a multiple of the number of layers. Enough layers remove this floor; more width only shrinks a separate Barron-space approximation term. Experiments with 54 trained Transformers on combinatorial search (Countdown) support these bounds. Appendix experiments measure structural shifts in public agent, embodied and robot trajectories, shift all position indices of 33 public language models, and split deeply nested tasks into separate model calls.

1 Introduction

The transition of large language models (LLMs) from associative pattern matching to explicit, multi-step reasoning has been driven first by prompting techniques [Wei et al., 2022, Yao et al., 2023] and, increasingly, by reinforcement learning and other post-training methods [Lightman et al., 2024, OpenAI, 2024, DeepSeek-AI, 2025, Lambert et al., 2024]. In formal logical domains, such as combinatorial planning and mathematical problem-solving, these intermediate reasoning trajectories correspond to traversals over an implicit algorithmic search tree. While the empirical scaling properties of reasoning are widely documented, theory lags behind. We ask: *Under what architectural conditions can an autoregressive model reliably generalize its reasoning capabilities across structural domain shifts?* When a model trained on shallow search steps is evaluated on deeper, more sequential out-of-distribution (OOD) reasoning spaces, standard sequence-level log-likelihoods and discrete zero-one

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40th Conference on Neural Information Processing Systems (NeurIPS 2026).

losses cannot measure the structural shift between training and test data, leaving this question largely unanswered.

Theoretical and empirical analyses of LLM reasoning remain fragmented. Complexity bounds for Transformers (TC^0) establish limitations on inherently sequential reasoning but abstract away training data distributions [Hahn, 2020, Merrill and Sabharwal, 2023, Feng et al., 2023]. Studies on length extrapolation evaluate mechanisms like Absolute Positional Encoding (APE) and Rotary Position Embedding (RoPE) [Kazemnejad et al., 2023, Ruoss et al., 2023], yet rarely do so while jointly bounding computational capacity. Data-centric analyses emphasize distributional shifts but treat the architecture as an analytic black box [Zhao et al., 2026, Prystawski et al., 2023]. No existing framework bridges discrete circuit complexity with continuous distribution shifts.

Our main contribution is a measure-theoretic framework that unifies four variables in LLM reasoning: discrete circuit complexity (layer depth L), representation capacity (hidden width m), architectural priors (positional encodings), and continuous data distributions (measure shifts). We formulate the evaluation of reasoning as an optimal transport problem. We project each discrete, variable-length reasoning trajectory into a continuous metric space whose twelve coordinates are structural statistics of the search, such as tree depth and backtracking rate; any reasoning dataset then becomes a pushforward probability measure, and the discrepancy between two reasoning tasks is quantified by the Wasserstein-1 distance between their measures.

We introduce an ϵ-relaxed structural sufficiency assumption that connects this continuous domain shift with the discrete expected risk. With the expected zero-one loss approximated by a Lipschitz continuous structural risk function over our metric space, we invoke Kantorovich duality and bound the target generalization risk by three terms: the optimal transport cost (Wasserstein domain shift), the induced Lipschitz constant of the architecture ($K_{f,\mathrm{PE}}$), and its functional approximation bounds (C_{width} and C_{depth}). Because the resulting lower bound is the maximum of these terms, exceeding any single capacity limit induces an $\Omega(1)$ expected risk. From this unified bound, we establish two architectural results:

First, we analyze translation equivariance and the Lipschitz gap by contrasting the two main families of sequence position modeling: absolute and relative encodings [Zhao et al., 2024]. We select standard Absolute Positional Encoding (APE) as the baseline for position-dependent attention kernels, and Rotary Position Embedding (RoPE) [Su et al., 2024] as the representative of a broader class of relative, translation-equivariant positional schemes. Generalization across positions requires the attention mechanism to remain invariant to global sequence shifts, as locally consistent search subtrajectories occur at arbitrary absolute indices. We prove that APE fails to preserve this symmetry; shifting a trajectory's absolute positions causes the local attention kernel to change almost surely. Under the autoregressive sensitivity of exact-match search (Assumption 2), this coordinate disruption yields a structural Lipschitz constant $K_{f,\mathrm{APE}} = \Omega(1)$ and an $\Omega(1)$ expected OOD risk whenever the target measure places non-vanishing mass on the disrupted structures (Theorem 1, Corollary 1). We show that RoPE instead preserves sequence equivariance and acts as a continuous structural regularizer, averting this representation collapse.

Second, building upon the TC^0 limits of Transformers [Merrill and Sabharwal, 2023], we derive a depth-dependent risk lower bound. Mapping the grammar of sequential backtracking to a Dyck-k language links continuous measure shifts with discrete circuit complexity. Because self-attention cannot model hierarchical structure unless the number of layers or heads grows with the input length [Hahn, 2020], we model the fixed physical layer depth L of TC^0 Transformers as an algorithmic truncation boundary (Assumption 5). Scaling physical depth is necessary to avert an $\Omega(1)$ expected risk; scaling representation width m cannot, as it only shrinks a separate Barron-space term of order $m^{-1/2}$ (Lemma 4; cf. the $\mathcal{O}(m^{-1/2})$ upper rates of Barron [1993] and Makovoz [1996]).

We test our framework on 54 trained models (crossing three training measures, Small/Medium/Large scales, Deep/Wide/Balanced aspect ratios, and APE/RoPE) against controlled target distributions. We synthesize distinct trajectory measures, isolating behaviors such as lateral state expansions, frequent recursive backtracking, and position-shifted compositional mixtures. The empirical results support our measure-theoretic bounds: (1) On the DFS evaluation set, all 54 models lie below the TC^0 accuracy ceiling motivated by Lemma 5 and calibrated on these models (Appendix G.3); doubling the width at fixed depth changes the best accuracy by under one point. (2) Scaling parameter counts via representation width plateaus or collapses: under APE, Deep models beat Wide ones in

all 18 Medium- and Large-scale comparisons (by up to 11.25 points), a gap that RoPE closes to under one point. (3) Under large coordinate shifts induced by mixture distributions, scaling Wide APE models triggers an $\Omega(1)$ representation collapse (0.4215 accuracy at Large scale), while RoPE at the same depth and width restores stability (0.5425). (4) RoPE outperforms APE in 25 of 27 matched model pairs (one-sided sign test $p < 10^{-5}$; 71 of 81 comparisons across evaluation sets), including all 18 Medium- and Large-scale pairs, with a gap that grows with scale. Overall, our framework casts LLM reasoning as structural regression over a continuous metric space.

2 Preliminaries and Measure-Theoretic Formulation

To analyze generalization on search-based reasoning tasks, we formalize autoregressive generation and map discrete, variable-length trajectories into a continuous metric space, where optimal transport quantifies domain shifts. Detailed measure-theoretic constructions and full proofs are deferred to Appendices A and B.

2.1 The Measurable Space of Search Trajectories

Let $\mathcal{V}$ be a finite discrete vocabulary. The ambient space of bounded-length sequences up to a maximum physical length $T_{\max} \in \mathbb{N}^+$ is defined as $\mathcal{Z}_{\text{all}} = \bigcup_{t=1}^{T_{\max}} \mathcal{V}^t$. We define the search trajectory space $\mathcal{Z} \subset \mathcal{Z}_{\text{all}}$ as the subset of sequences satisfying a valid autoregressive search grammar Γ, i.e., the rules by which a search tree is serialized into tokens (expand a node, evaluate a child, prune a dead end, backtrack to a pending node); its transition alphabet is formalized in Appendix D.2.1. Every trajectory $z \in \mathcal{Z}$ admits a unique decomposition via the string concatenation operator $\oplus$: $z = x_{\text{init}}^{(z)} \oplus y_{\text{search}}$, representing the initial problem state and the generated search trace.

Given finite $|\mathcal{V}|$ and $T_{\max}$, the set $\mathcal{Z}$ is finite. We equip $\mathcal{Z}$ with the discrete σ-algebra $\Sigma_{\mathcal{Z}} := 2^{\mathcal{Z}}$, forming a complete measurable space with source and target probability measures $\mu_S, \mu_T \in \mathcal{P}(\mathcal{Z})$.

Let $\mathcal{F}_{L,m,\text{PE}}$ denote the hypothesis class of autoregressive Transformer functions characterized by physical layer depth L, hidden dimension m, and positional encoding scheme $\text{PE} \in \{\text{APE}, \text{RoPE}\}$. A model $f \in \mathcal{F}_{L,m,\text{PE}}$ is a deterministic mapping $f : \mathcal{V}^* \to \mathcal{Z}_{\text{all}}$. Let $S_z : \mathcal{Z}_{\text{all}} \to \mathbb{R}$ be a deterministic scoring function parameterized by the target state of the reference trajectory z, evaluating the formal validity of the generated output. We define the zero-one loss function $\ell : \mathcal{F}_{L,m,\text{PE}} \times \mathcal{Z} \to \{0, 1\}$ as:

$$\ell(f, z) = 1 - \mathbb{I}_{(0,\infty)} \left(S_z \left(f(x_{\text{init}}^{(z)}) \right) \right) \tag{1}$$

For any measure $\mu \in \mathcal{P}(\mathcal{Z})$, the expected risk of f is defined as the Lebesgue integral over the discrete space: $\mathcal{R}_\mu(f) = \int_{\mathcal{Z}} \ell(f, z)\, d\mu(z)$.

2.2 Structural Projection and Pushforward Measures

Computing distances directly over discrete sequences obscures search behavior. Thus, we map trajectories into a continuous metric space $\mathcal{X} \subset \mathbb{R}^{12}$ equipped with the Euclidean metric $d_{\mathcal{X}}(\boldsymbol{u}, \boldsymbol{v}) = \|\boldsymbol{u} - \boldsymbol{v}\|_2$. As volumetric coordinates (e.g., χ_4) scale with sequence capacity $T_{\max}$, the geometric diameter $\text{diam}(\mathcal{X}) = \Theta(T_{\max})$. Thus, for any finite $T_{\max}$, $\mathcal{X}$ is a bounded, locally compact subset.

Definition 1 (Structural Projection Operator Φ)**.** *We define a deterministic, Borel measurable projection operator $\Phi : \mathcal{Z} \to \mathcal{X}$, mapping $z \mapsto \boldsymbol{\chi} = [\chi_1, \ldots, \chi_{12}]^\top$. This operator provides a summary of the graph-traversal dynamics, encoding maximum tree depth (χ_1), total search volume (χ_4), directional state transitions (e.g., child expansion χ_5, parent backtracking χ_6, sibling lateral shifts χ_7), and pruning frequencies. In plain terms, χ_4 is the number of node references in the trace, while χ_5, χ_6, χ_7 are the fractions of consecutive nodes that go deeper, go back up, or stay at the same depth of the tree; formal definitions of all twelve coordinates are given in Appendix A.1.*

Pushing μ_S and μ_T forward through Φ gives the *structural pushforward measures* $\nu_S := \Phi_{\#}\mu_S$ and $\nu_T := \Phi_{\#}\mu_T$ in $\mathcal{P}(\mathcal{X})$. The structural measure shift is quantified by the Wasserstein-1 (Kantorovich–Rubinstein) distance over the coupled metric space [Villani, 2009]:

$$W_1(\nu_S, \nu_T) = \inf_{\pi \in \Pi(\nu_S, \nu_T)} \int_{\mathcal{X} \times \mathcal{X}} d_{\mathcal{X}}(\boldsymbol{u}, \boldsymbol{v})\, d\pi(\boldsymbol{u}, \boldsymbol{v}) \tag{2}$$

where $\Pi(\nu_S, \nu_T)$ denotes the set of all joint probability measures on $\mathcal{X} \times \mathcal{X}$ with marginals ν_S and ν_T. Because $\mathcal{X}$ is bounded for finite $T_{\max}$, the integral converges absolutely, yielding $W_1 \leq \mathcal{O}(T_{\max})$.

Why a structural projection. Token-level likelihoods cannot express how far apart two search strategies are: a shallow breadth-first trace and a deep backtracking trace may have similar likelihood yet differ entirely in how they route through the tree. Mapping each trajectory to its traversal statistics turns every reasoning task into a probability measure on $\mathcal{X}$, so the task shift becomes a single transport cost $W_1(\nu_S, \nu_T)$, while the architecture enters only through the Lipschitz constant of its risk function over $\mathcal{X}$ (Lemma 1). This separation of data shift from architectural sensitivity is what the rest of the paper uses.

2.3 Error Decomposition via Kantorovich Duality

We relate the continuous domain shift $W_1(\nu_S, \nu_T)$ to the discrete expected risk $\mathcal{R}_\mu(f)$ through a sufficiency assumption that decouples the architecture's capacity from the empirical data distribution.

Assumption 1 (ϵ-Relaxed Structural Sufficiency)**.** *Let $\epsilon > 0$ be an irreducible approximation error. We assume there exists a measurable risk function $\mathcal{L}_f : \mathcal{X} \to [0, 1]$. For any trajectory measure $\mu \in \{\mu_S, \mu_T\}$, the absolute difference between the true expected risk on $\mathcal{Z}$ and the expected risk integrated over $\Phi_{\#}\mu$ on $\mathcal{X}$ is bounded by ϵ:*

$$\left| \mathcal{R}_\mu(f) - \int_{\mathcal{X}} \mathcal{L}_f(\boldsymbol{\chi})\, d(\Phi_{\#}\mu)(\boldsymbol{\chi}) \right| \leq \epsilon \tag{3}$$

This assumption states that the structural projection $\Phi(z)$ captures the task-relevant components of the search trajectory up to a residual ϵ. It targets formal combinatorial planning and algorithmic search, where the structure of a problem's search (e.g., tree depth and backtracking) summarizes its difficulty. The out-of-distribution generalization gap is then bounded by the Lipschitz continuity of the neural architecture (Lemma 1). Appendix J checks the assumption on two trained models that differ in scale, training measure and positional encoding: a gradient boosting classifier on $\boldsymbol{\chi}$ alone predicts whether the model solves a problem with out-of-fold ROC-AUC 0.93 and McFadden pseudo-R^2 0.48–0.51 (token length alone: 0.44–0.46; the problem alone: 0.40–0.41), and its ROC-AUC exceeds that of either baseline in every fold.

Lemma 1 (ϵ-Relaxed Error Decomposition)**.** *Assume the induced risk function $\mathcal{L}_f(\boldsymbol{\chi})$ is Lipschitz continuous on $(\mathcal{X}, d_{\mathcal{X}})$ with minimal Lipschitz constant $K_f = \sup_{\boldsymbol{u} \neq \boldsymbol{v}} |\mathcal{L}_f(\boldsymbol{u}) - \mathcal{L}_f(\boldsymbol{v})| / d_{\mathcal{X}}(\boldsymbol{u}, \boldsymbol{v})$. Then the generalization error gap is bounded by:*

$$|\mathcal{R}_{\mu_T}(f) - \mathcal{R}_{\mu_S}(f)| \leq K_f \cdot W_1(\nu_S, \nu_T) + 2\epsilon \tag{4}$$

Proof Sketch. Applying the triangle inequality alongside Assumption 1 bounds the risk difference $|\Delta\mathcal{R}|$ by $|\int \mathcal{L}_f d\nu_T - \int \mathcal{L}_f d\nu_S| + 2\epsilon$. Normalizing the structural risk function as $g = \mathcal{L}_f / K_f$ yields $\|g\|_{\mathrm{Lip}} \leq 1$. Invoking the Kantorovich–Rubinstein duality [Villani, 2009], the normalized integral difference $|\int g\, d\nu_T - \int g\, d\nu_S|$ is bounded by $W_1(\nu_S, \nu_T)$. Scaling the duality bound by K_f yields the claim. □

3 Theoretical Limits of Architectural Choices

By Lemma 1, the out-of-distribution generalization gap is bounded by the Lipschitz constant K_f and the approximation bounds of $\mathcal{L}_f$. We now formalize how architectural priors govern these bounds along two computational axes of the autoregressive Transformer: the positional encoding scheme (APE vs. RoPE) and the aspect ratio of the computational graph (Deep vs. Wide).

3.1 Sequence Equivariance and the Lipschitz Gap

Algorithmic search trajectories are modular. A locally consistent reasoning sub-trajectory (e.g., a specific subtree traversal) can occur at arbitrary absolute sequence indices, depending on the volume of preceding graph explorations. To quantify a model's sensitivity to these shifts, we examine the behavior of the attention kernel under sequence translations.

Definition 2 (Minimal Sequence Translation Operator $\mathcal{T}_k$)**.** *Let $z \in \mathcal{Z}$. We define the translation operator $\mathcal{T}_k : \mathcal{Z} \to \mathcal{Z}$ for $k \in \mathbb{N}^+$ as the operation that inserts one structurally inert node whose textual realization contains exactly k tokens immediately before a local reasoning sub-trajectory ω.*

Under Φ, this increments the search volume ($\chi'_4 = \chi_4 + 1$), while every token in the subsequent ω undergoes an absolute index shift of $+k$. Hence the distance in $\mathcal{X}$ is

$$d_{\mathcal{X}}(\Phi(z), \Phi(\mathcal{T}_k(z))) = \left(1^2 + \sum_{m \neq 4} (\Delta\chi_m)^2\right)^{1/2} =: \delta \leq C_\delta \tag{5}$$

where C_δ is independent of $T_{\max}$.

For a model to generalize across diverse tree structures, its local computational graph must remain invariant under $\mathcal{T}_k$. We evaluate the pre-softmax attention score $A_{a,b}$ between two tokens in ω originally at indices a and b, now shifted to $a + k$ and $b + k$.

Lemma 2 (APE Absolute Coordinate Sensitivity)**.** *Assume the network parameters (projection matrices $\mathbf{W}_Q, \mathbf{W}_K$ and absolute embeddings $\mathbf{p}_t$) are generically parameterized, and the projection matrices satisfy $\mathbf{W}_Q^\top \mathbf{W}_K \neq \mathbf{0}$. Under the operator $\mathcal{T}_k$ ($k > 0$), the local computational attention kernel of an APE model differs almost surely: $\Delta A^{\mathrm{APE}} = |\tilde{A}^{\mathrm{APE}}_{a,b} - A^{\mathrm{APE}}_{a,b}| > 0$.*

Proof Sketch. Expanding the bilinear forms of the APE attention kernel cancels the position-independent terms. The residual difference ΔA^{APE} forms a nontrivial multivariate quadratic polynomial. By the properties of real algebraic geometry, the zero-locus of this polynomial defines a proper algebraic variety with Lebesgue measure zero in $\mathbb{R}^{2d}$. Consequently, generic parameters avoid this set, so exact cancellation does not occur. Rotary Position Embedding (RoPE) instead relies on orthogonal rotation matrices, yielding $\Delta A^{\mathrm{RoPE}} = 0$. See Appendix C. □

Assumption 2 (Autoregressive Sensitivity via Markov Compounding)**.** *We assume the base search-based reasoning task exhibits an intrinsic global smoothness, yielding a baseline structural Lipschitz constant $\mathcal{O}(1/T_{\max})$. Generating a formal search trajectory constitutes a Markov decision process over a directed acyclic graph. We assume that a local attention-kernel coordinate disruption ($\Delta A > 0$) at a critical routing step sends the autoregressive process into an incorrect subtree. Due to exact-match evaluation, this local deviation precludes reaching the target state, compounding the localized error into a global trajectory failure. This bounds the expected structural risk variation below by the uniform error probability: $\Delta\mathcal{L} = |\mathcal{L}_f(\boldsymbol{\chi}') - \mathcal{L}_f(\boldsymbol{\chi})| \geq 1 - 1/|\mathcal{V}| =: \gamma > 0$.*

Theorem 1 (The Lipschitz Gap)**.** *Under Assumption 2, for a sufficiently large sequence capacity $T_{\max}$, the Lipschitz constant of the induced risk function for APE is asymptotically larger than that of RoPE: $K_{f,\mathrm{RoPE}} = \mathcal{O}(1/T_{\max}) \ll \Omega(1) \leq K_{f,\mathrm{APE}}$.*

Proof Sketch. The Lipschitz constant is bounded below by the evaluation at the specific perturbed pair: $K_{f,\mathrm{APE}} \geq |\Delta\mathcal{L}|/\delta \geq \gamma/C_\delta$. Because γ and C_δ are positive constants independent of $T_{\max}$, it follows that $K_{f,\mathrm{APE}} = \Omega(1)$. By preserving relative sequence distances, RoPE maintains the local dependency structure without introducing artificial discontinuities, inheriting the task's intrinsic baseline smoothness $\mathcal{O}(1/T_{\max})$. □

Causal reading. The causation runs from the architecture to the risk. Because $\mathcal{L}_f$ is indexed by the model f, the positional encoding shapes the risk function over $\mathcal{X}$; K_f measures how much this risk can change under a small structural shift δ. Under $\mathcal{T}_k$, which moves $\boldsymbol{\chi}$ by $\delta \leq C_\delta$ and every token of ω by k positions, the APE attention kernel changes (Lemma 2), and in exact-match search this local change diverts greedy routing into a wrong subtree (Assumption 2). A shift of δ in $\mathcal{X}$ thus produces a jump γ in risk, i.e., $K_{f,\mathrm{APE}} = \Omega(1)$. Through Lemma 1, the guarantee scales with $K_f W_1$, and RoPE keeps K_f at $\mathcal{O}(1/T_{\max})$.

3.2 Capacity Limits: The TC^0 Algorithmic Bottleneck

Beyond sequence equivariance, graph-traversal dynamics impose distinct computational complexity constraints on the hypothesis space $\mathcal{F}_{L,m,\mathrm{PE}}$.

In shallow, lateral state expansions, maintaining mutually independent states allows the evaluation process to be efficiently parallelized across attention heads in a wide network. By contrast, deep recursive backtracking requires a Last-In-First-Out (LIFO) stack mechanism. To execute a backtracking transition, the model must implicitly identify and resolve the most recently suspended, non-adjacent sibling node across a flattened 1D sequence history.

Lemma 3 (Surjective Monoid Homomorphism to Dyck-k)**.** *Let k denote the maximum branching factor of the implicit search tree. Defining the traversal alphabet $\Sigma_{\text{search}} = \{\delta_{\text{down}}^{(1)}, \dots, \delta_{\text{down}}^{(k)}\} \cup \{\delta_{\text{up}}^{(1)}, \dots, \delta_{\text{up}}^{(k)}\} \cup \{\omega_{\text{eval}}\}$, there exists a surjective monoid homomorphism $\psi : \Sigma_{\text{search}}^* \to \Sigma_{\text{Dyck}}^*$ mapping child expansions to left parentheses $(_c$ and backtracking steps to right parentheses $)_c$. Every valid search trajectory projects to a prefix of a word in $\mathcal{D}_k$, where the maximum tree depth χ_1 determines the maximum parenthesis nesting depth.*

A Transformer computing L sequential layers under bounded numerical precision belongs to TC^0 [Merrill and Sabharwal, 2023], and self-attention cannot model hierarchical structure unless the number of layers or heads grows with the input length [Hahn, 2020]; we therefore model resolving deeper nesting within one forward pass as requiring more layers (Assumption 5, Appendix D.2.2). We formalize this limitation into two independent capacity bounds on the expected risk.

Let $\mathcal{L}^*(\boldsymbol{\chi}) = \inf_h \mathbb{E}_{z\sim\mu_T}[\ell(h, z) \mid \Phi(z) = \boldsymbol{\chi}]$ denote the Bayes-optimal structural risk function. Lemma 4 transfers the $L_1(\nu_T)$ approximation error in $\mathcal{H}_{\text{Barron}}$ to the discrete zero-one risk via the Bayes optimality of $\mathcal{L}^*$.

Lemma 4 (Width Capacity Bound via Barron Spaces)**.** *Under Assumptions 1 and 4 (in particular $\mathcal{L}^* \in \mathcal{H}_{\text{Barron}}$), any autoregressive Transformer hypothesis bounded by an internal representation width m satisfies the lower bound: $\mathcal{R}_{\mu_T}(f) \geq \frac{C_{\text{width}}}{\sqrt{m}} - \epsilon$.*

Proof Sketch. While the Transformer operates on token-level embeddings, correctly routing recursive transitions requires its internal representations to implicitly track global tree states. Thus, the hidden width m (which also fixes the FFN width) acts as an information-theoretic upper bound on the capacity of $\mathcal{L}_f$. Since $\mathcal{L}^* \geq 0$ is the pointwise Bayes risk, $\int \mathcal{L}^* \, d\nu_T \leq \mathcal{R}_{\mu_T}(f)$; together with Assumption 1 and $|\mathcal{L}_f - \mathcal{L}^*| \leq \mathcal{L}_f + \mathcal{L}^*$, this gives $2\mathcal{R}_{\mu_T}(f) \geq \int |\mathcal{L}_f - \mathcal{L}^*| \, d\nu_T - \epsilon$. Classical Barron-space results supply the upper rate: some width-m network attains an $\mathcal{O}(m^{-1/2})$ error [Barron, 1993, Makovoz, 1996]. The lemma needs the matching lower bound, which Assumption 4(iii) (Appendix D.1) states as a minimax-hardness condition: no width-m hypothesis approximates $\mathcal{L}^*$ faster than order $m^{-1/2}$. With the inequality above, this yields the claim. □

While the external autoregressive generation linearly expands the global causal graph length, the model must execute implicit heuristic routing evaluations at each discrete step that are not serialized in the output. Resolving non-adjacent sibling dependencies across a flattened history relies entirely on the fixed physical depth L of a single forward pass.

Motivated by these depth limits, we introduce a task-dependent truncation constant $\alpha_{\text{circ}} > 0$ and model the recoverable structural depth of a depth-L architecture by the threshold $\chi_1 \leq \alpha_{\text{circ}} L$.

Lemma 5 (Implicit Depth Bottleneck Bound)**.** *Under Assumptions 1 and 5, defining the Borel measurable indicator function $P_{\text{implicit}}(\boldsymbol{\chi}; L, \alpha_{\text{circ}}) = \mathbb{I}_{(\alpha_{\text{circ}} L, \infty)}(\chi_1)$, the expected target risk is bounded below by: $\mathcal{R}_{\mu_T}(f) \geq C_{\text{depth}} \int_{\mathcal{X}} P_{\text{implicit}} \, d\nu_T - \epsilon$, where $C_{\text{depth}} = 1 - \frac{1}{|\mathcal{V}|} \in (0, 1)$.*

4 Synthesis of the Main Bound

Theorem 2 (The Generalization Bound)**.** *Under Assumptions 1, 4 and 5 and a Lipschitz $\mathcal{L}_f$, for any autoregressive Transformer $f \in \mathcal{F}_{L,m,\text{PE}}$, the target risk $\mathcal{R}_{\mu_T}(f)$ is bounded below by the maximum of three independent lower bounds, and from above by its source risk and domain shift penalty:*

$$\textit{(LB)} \quad \mathcal{R}_{\mu_T}(f) \geq \max\Big\{ \frac{C_{\text{width}}}{\sqrt{m}} - \epsilon,\ C_{\text{depth}} \int_{\mathcal{X}} P_{\text{implicit}}\, d\nu_T(\boldsymbol{\chi}) - \epsilon, \\ \mathcal{R}_{\mu_S}(f) - K_{f,\text{PE}} \cdot W_1(\nu_S, \nu_T) - 2\epsilon \Big\} \tag{6}$$

$$\textit{(UB)} \quad \mathcal{R}_{\mu_T}(f) \leq \mathcal{R}_{\mu_S}(f) + K_{f,\text{PE}} \cdot W_1(\nu_S, \nu_T) + 2\epsilon \tag{7}$$

Proof Sketch. Applying Kantorovich duality to the ϵ-relaxed structural risk (Lemma 1), and intersecting this with the implicit-depth bound (Lemma 5) and the Barron-space width bound (Lemma 4), isolates the respective bounds. See Appendix E. □

Non-substitutable lower bounds. Each term of (LB) is a valid lower bound arising from a different mechanism (width, depth, measure shift), so their maximum is also a lower bound; a sum would require the corresponding failure events to be disjoint and could exceed the trivial bound $\mathcal{R} \leq 1$. Every term thus acts as a necessary condition: if a target tree requires tree depth $\chi_1 = 20$ but $L = 6$, the depth term remains $\Omega(1)$ however large m is, even if RoPE removes the shift penalty (Appendix I).

Corollary 1 (The Divergent OOD Guarantees of APE vs. RoPE)**.** *Let $\mathcal{R}_{\mu_S}(f) = \epsilon_{\text{train}} \approx 0$ and $W_1(\nu_S, \nu_T) = \Omega(T_{\max})$. RoPE preserves equivariance: with $K_{f,\text{RoPE}} \leq C_{\text{smooth}}/T_{\max}$ (Theorem 1), its expected generalization error degrades at most proportionally to the normalized shift, $\mathcal{R}_{\mu_T}(f_{\text{RoPE}}) \leq \epsilon_{\text{train}} + C_{\text{smooth}} \cdot (W_1/T_{\max}) + 2\epsilon$. APE, however, suffers coordinate disruption, yielding an $\Omega(1)$ expected risk: $\mathcal{R}_{\mu_T}(f_{\text{APE}}) \geq \gamma \cdot p_{\text{shift}} - \epsilon$, where $p_{\text{shift}} > 0$ is the target mass of structures on which its routing fails ($\mathcal{L}_{f_{\text{APE}}} \geq \gamma$; Assumption 2), precluding any generalization guarantee.*

Corollary 2 (The "Deep is Better" Law for Search Reasoning)**.** *Let the target measure ν_T place probability mass on trajectories satisfying the tree depth threshold $\chi_1 > \alpha_{\text{circ}} L$. Scaling the discrete physical depth $L \geq \alpha_{\text{circ}}^{-1} \sup_{\boldsymbol{\chi}} \chi_1$ eliminates the $\Omega(1)$ lower bound, establishing a dominant scaling trajectory over the representation width m, which plateaus as $\lim_{m\to\infty}(C_{\text{width}}/\sqrt{m} - \epsilon) = -\epsilon \leq 0$.*

5 Empirical Verification

To test architectural predictions of Theorem 2, we compare architectures under controlled configurations by decoupling layer depth L, width m, and positional encodings, while holding the core parameter count $12Ld^2$ nearly constant per scale. Motivated by depth-separation theorems [Eldan and Shamir, 2016, Levine et al., 2021] and TC^0 bounds [Merrill and Sabharwal, 2023], we design a grid of nine Transformer architectures crossing three parameter scales (Small, Medium, Large) and three aspect ratios (Deep, Wide, Balanced).

To vary the Lipschitz constant $K_{f,\text{PE}}$, we cross every architecture with APE and RoPE [Su et al., 2024]. While empirical literature highlights their role in length extrapolation, our framework casts them in structural terms: APE imposes an absolute coordinate system that translations disrupt, whereas RoPE injects relative equivariance to preserve continuous structural regularities. Table 1 summarizes these models.

To instantiate the grammar Γ and probability measures (μ_S, μ_T), we use the Stream of Search (SoS) framework [Gandhi et al., 2024] as a discrete trajectory simulator. SoS lets us fix the search procedure that generates each measure, providing a controlled testbed. We synthesize 500,000 ground-truth trajectories per measure on a combinatorial planning task (the Countdown game), in which a set of input numbers must be combined with arithmetic operations to reach a target (e.g., reaching 10 from $[2, 3, 5]$); SoS serializes the resulting search tree, including dead ends and backtracking, into one token sequence (worked example in Appendix K.4). Exact-match verification on this formal task isolates structural routing errors from semantic noise, as required to test the depth and Lipschitz terms of Theorem 2. By varying the simulator's heuristic evaluation strategies, we generate distinct distributions. The μ_{BFS} measure is synthesized via queue-based state expansions, generating lateral traversal traces. The μ_{DFS} measure is synthesized via recursive stack expansions, producing traces with nearly four times the backtracking rate χ_6 of μ_{BFS}. The μ_{MIXED} distribution is constructed as a $0.5\mu_{BFS} + 0.5\mu_{DFS}$ mixture (with the search heuristic of each component drawn at random)

Table 1: Summary of the 54 Transformer Architectural Configurations Evaluated

Scale / Shape	**Layers** (L)	**Width** (d)	**Heads** (h)	**Core Params**†	**Total Params**‡
Small Deep	12	192	4	5.3 M	24.8 M
Small Balanced	6	256	4	4.7 M	30.7 M
Small Wide	3	384	4	5.3 M	44.3 M
Medium Deep	24	512	8	75.5 M	127.6 M
Medium Balanced	12	768	12	84.9 M	163.0 M
Medium Wide	6	1024	16	75.5 M	179.5 M
Large Deep	48	768	12	339.7 M	418.1 M
Large Balanced*	24	1024	16	302.0 M	406.2 M
Large Wide	12	1536	24	339.7 M	495.9 M

Note: All architectures crossed with PE $\in \{$*APE, RoPE*$\}$ *and Training Data* $\in \{$*BFS, DFS, MIXED*$\}$.

†*Core Params* = $12Ld^2$*, isolating the architectural capacity independent of vocabulary.*

‡*Total Params* = $12Ld^2 + 10Ld + 2d + 2Vd + 1024d$ *(*$V = 50{,}257$*), i.e., GPT-Neo with untied embeddings, an upper bound (APE models tie embeddings; RoPE models have no position table).*

**Adjusted* $d = 1024$ *to match the computational capacity of its scale group.*

to test architectures under large position shifts. Under exact-match scoring, the zero-one loss credits only outputs that reach the target with valid arithmetic.

Structural Optimal Transport Estimation. To measure the transport cost in the Kantorovich bounds, we estimate the Wasserstein-1 distance between structural pushforward measures by solving the exact (unregularized) optimal transport problem between empirical measures with POT [Flamary et al., 2021]. Using $N = 2{,}000$ uniformly sampled trajectories per distribution and 10 independent subsamples, we construct empirical structural measures on $\mathcal{X}$. Following global standardization, we obtain $\widehat{W}_1(\hat{\nu}_{BFS}, \hat{\nu}_{MIXED}) = 1.38 \pm 0.05$ and $\widehat{W}_1(\hat{\nu}_{BFS}, \hat{\nu}_{DFS}) = 2.97 \pm 0.05$, against a within-measure sampling baseline of 0.19 ± 0.06; the mixture thus sits at 46% of the BFS-to-DFS shift. Estimation procedures, cost-matrix construction, normalization details, and the variation across repeated subsamples are reported in Appendix G. On this standardized scale, both shifts exceed the within-measure baseline at least sevenfold.

5.1 Positional Encodings as Structural Regularizers

Turning to positional encodings as regularizers, we examine the Lipschitz gap established in Theorem 1. As predicted by our bounds, APE yields no generalization guarantee under sequence translations. When trained on the μ_{MIXED} distribution, the model alternates between BFS- and DFS-style traversals, subjecting the attention kernel to large absolute sequence index shifts. The Large Wide MIXED APE model reaches only 0.4215 accuracy on the MIXED evaluation set, which we attribute to the coordinate disruption of Lemma 2.

In contrast, preserving local sequence equivariance allows the Large Wide MIXED RoPE model to reach 0.5425, an absolute gain of over 12 points. The two models share depth $L = 12$ and width $m = 1536$ and differ principally in the positional encoding. Across the full grid, RoPE outperforms APE in 71 of 81 paired comparisons (same scale, shape, training measure and evaluation set) and in 25 of 27 model pairs averaged over the evaluation sets (one-sided sign test $p < 10^{-5}$), and the advantage grows with scale (mean gap $+0.6$, $+1.2$ and $+3.3$ points for Small, Medium and Large; 27 of 27 comparisons at Large scale); Figure 1D shows the same ordering, on average, on every evaluation set for the BFS-trained models. This is consistent with the Lipschitz gap of Theorem 1, under which RoPE is both a length extrapolator and a regularizer over the continuous metric space.

5.2 The Anisotropy of Parameter Scaling and TC^0 Bottlenecks

Our bounds imply an anisotropy of parameter scaling: overcoming the TC^0 complexity bottleneck requires increasing the physical layer depth L (Lemma 5), while widening the representation dimension m plateaus within Barron spaces (Lemma 4). Under APE, Deep beats Wide in all 18 Medium- and Large-scale comparisons; at equal core parameters, the Large Deep MIXED APE

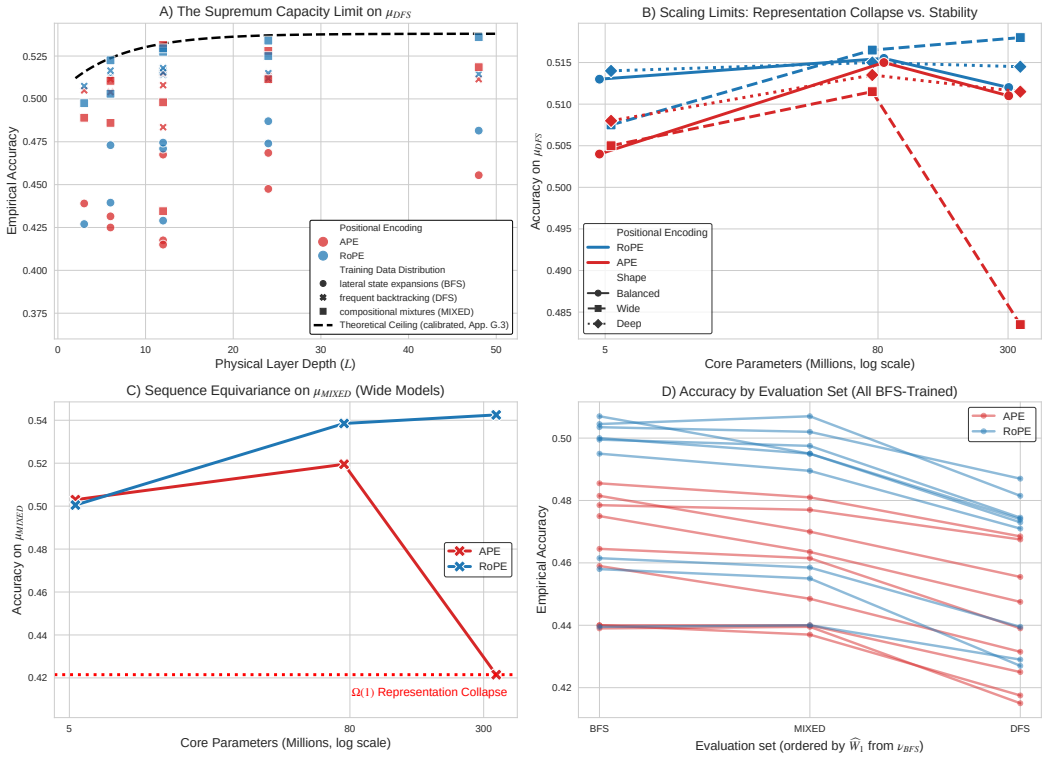


Figure 1: Validation of capacity bounds. (A) Supremum capacity limit on μ_{DFS}, showing empirical accuracy enveloped by the calibrated upper bound curve (Appendix G.3). (B) Scaling limits: width plateaus and, under APE, collapses at Large scale (Wide 0.4835 vs. Deep 0.5115); with RoPE, Wide matches Deep. (C) Sequence equivariance on μ_{MIXED}, where scaling Wide APE architectures triggers representation collapse, while RoPE acts as a regularizer. (D) All BFS-trained models on the three evaluation sets, ordered by Wasserstein distance from ν_{BFS}; RoPE outperforms APE on average on each set (in 23 of 27 comparisons). The sets share one problem distribution, so the decline toward μ_{DFS} reflects the difficulty of their problem samples (Appendix K.5).

model (0.5330) exceeds its Wide counterpart (0.4215) by 11.15 points. Twelve layers suffice on this task: at the same $L = 12$, Wide RoPE reaches 0.5425 and the half-width Medium Balanced MIXED APE 0.5360, so the extra depth compensates for a collapse specific to Wide APE (Appendix H). With RoPE, this collapse disappears and Wide matches Deep (Figure 1B): at Large scale the gap stays below one point in all nine comparisons, against 2.6–11.25 points under APE. This indicates that translation equivariance allows the architecture to approach its theoretical capacity without the coordinate disruption that limits APE models, illustrating the interaction between Lipschitz smoothness and structural capacity.

5.3 The Supremum Bound

Finally, we validate the unified bound (Theorem 2), which implies that generalization is governed by a supremum operator over architectural bottlenecks and optimal transport costs. In Figure 1A, all models lie below the calibrated TC^0 ceiling, and doubling the width at fixed depth changes the best accuracy by under one point (Appendix G.3), confirming that width does not raise this ceiling.

6 Related Work

Circuit Complexity and Transformer Expressivity. Bounded-precision Transformers are restricted to TC^0 [Merrill and Sabharwal, 2023], and self-attention cannot model hierarchical structure such as Dyck languages unless the number of layers or heads grows with the input length [Hahn,

2020]. Depth-separation theorems further prove that bounded depth cannot be efficiently compensated by width [Eldan and Shamir, 2016, Levine et al., 2021]. While these works yield distribution-agnostic worst-case limits, our framework integrates TC^0 bottlenecks with optimal transport and Barron-space bounds [Barron, 1993, Makovoz, 1996], mapping discrete expressivity directly to expected risk under specific data measures.

Positional Encodings as Structural Regularizers. Positional encodings govern OOD length extrapolation [Zhao et al., 2024], evolving from absolute injections (APE) to relative mechanisms like ALiBi [Press et al., 2022] and to models without positional encoding (NoPE) [Kazemnejad et al., 2023]. Recent variants, including functional interpolation (FIRE) [Li et al., 2024], hierarchical encodings (BiPE) [He et al., 2024], and data-adaptive schemes (DAPE) [Zheng et al., 2024], report improved empirical generalization. Beyond theories of decaying attention, we cast shift-invariant encodings as structural regularizers. Relative encodings that share the translation-equivariant algebraic signature $\Delta A = 0$ under shifts inherit our generalized Lipschitz bound. Thus, Theorem 1 and Corollary 1 quantify why APE incurs an $\Omega(1)$ risk under structural shifts, whereas equivariant priors bound the Kantorovich penalty.

Optimal Transport and Reasoning Domain Shifts. Recent analyses conceptualize reasoning as search tree traversals [Gandhi et al., 2024] and relate its generalization to train-test distributional discrepancies [Prystawski et al., 2023, Zhao et al., 2026]. However, they treat neural architectures as black boxes. To address this, we draw on optimal transport (OT) domain adaptation, which bounds target risk via Wasserstein distances over static Euclidean features [Courty et al., 2017, Redko et al., 2017]. We extend this OT framework to *autoregressive sequence generation*. With our pushforward operator Φ, which projects variable-length trajectories into a continuous metric space, we couple Kantorovich dual bounds to Transformer circuit complexity and attention kernels, providing the first unified bound bridging continuous domain shifts with discrete architectural limits.

7 Conclusion and Limitations

We introduced a measure-theoretic framework quantifying the structural generalization of LLMs engaged in formal search-based reasoning. It maps discrete reasoning trajectories into a continuous metric space and decomposes the OOD generalization gap via Kantorovich duality. Under the stated assumptions, we proved two architectural results, and our experiments support their consequences: (1) APE suffers coordinate disruptions under sequence translations, resulting in an $\Omega(1)$ Lipschitz constant and an $\Omega(1)$ expected risk, whereas RoPE guarantees relative equivariance; and (2) fixed-depth Transformers incur a risk floor on Dyck-k recursive backtracking that scaling depth (L) removes and width (m) cannot bypass.

Limitations and Future Work. The ϵ-relaxed structural sufficiency assumption targets formal combinatorial search; we leave the estimation of ϵ for open-ended natural-language reasoning to future work. The twelve coordinates of Φ are one instantiation of the framework, and our three measures share one problem distribution; other projections, and target measures that also change the problems posed and thereby test the $K_f W_1$ term directly, are natural next steps (Appendix K.5). Appendix M derives from the bound what the coordinates of Φ must satisfy, measures structural shifts between public agent, embodied and robot trajectories, and tests two practical consequences of the theory: shifted position indices in serving, and splitting deeply nested tasks into separate model calls. Our TC^0 bottleneck concerns the forward-pass routing of a single autoregressive trajectory; how inference-time search such as Tree-of-Thoughts or Monte Carlo Tree Search reshapes the target measure ν_T is a direction for future theory.

Broader Impact. As the field moves toward inference-time scaling and search-based LLMs, our framework implies that translation equivariance, as in RoPE, is a necessary condition for stable inference-time scaling, given the large sequence translations ($T_{\max} \to \infty$) involved. Together with the necessity of depth, this result gives principled guidance for designing more efficient reasoning architectures.

Acknowledgments and Disclosure of Funding

We thank Qihuang Zhang (Department of Epidemiology, Biostatistics and Occupational Health, McGill University) for supporting this work.

This research was enabled in part by compute resources, software, and technical assistance provided by the Mila cluster.

Funding. Computing resources were provided by Mila – Quebec AI Institute. This work was supported by the Natural Sciences and Engineering Research Council of Canada (NSERC) through a Discovery Grant (RGPIN-2023-03854). No other third-party funding was received.

Competing interests. The authors declare no competing interests.

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A Formal Setup and Notation

This section collects the formal objects used throughout the appendix and fixes notation for the structural measure-theoretic formulation of search-based reasoning.

A.1 Trajectory Space and Structural Feature Space

Let $\mathcal{V}$ be a finite vocabulary and let $T_{\max} \in \mathbb{N}^+$ denote the maximum physical sequence length. The ambient space of bounded-length sequences is

$$\mathcal{Z}_{\text{all}} := \bigcup_{t=1}^{T_{\max}} \mathcal{V}^t. \tag{8}$$

Let Γ denote the autoregressive grammar of valid search trajectories. The trajectory space is

$$\mathcal{Z} := \{z \in \mathcal{Z}_{\text{all}} : z \text{ satisfies } \Gamma\}. \tag{9}$$

Since $|\mathcal{V}| < \infty$ and $T_{\max} < \infty$, the set $\mathcal{Z}$ is finite. We equip $\mathcal{Z}$ with the discrete σ-algebra

$$\Sigma_{\mathcal{Z}} := 2^{\mathcal{Z}}. \tag{10}$$

We consider source and target probability measures

$$\mu_S, \mu_T \in \mathcal{P}(\mathcal{Z}), \tag{11}$$

where $\mathcal{P}(\mathcal{Z})$ denotes the set of probability measures on $(\mathcal{Z}, \Sigma_{\mathcal{Z}})$.

Each trajectory $z \in \mathcal{Z}$ is written as

$$z = x_{\text{init}}^{(z)} \oplus y_{\text{search}}, \tag{12}$$

where $x_{\text{init}}^{(z)} \in \mathcal{V}^k$ is the initial problem state and $y_{\text{search}} \in \mathcal{V}^{t-k}$ is the generated search trace.

We next define the structural feature space. Let $\mathcal{X} \subset \mathbb{R}^{12}$ be the structural state space, equipped with the Euclidean metric

$$d_{\mathcal{X}}(\mathbf{u}, \mathbf{v}) := \|\mathbf{u} - \mathbf{v}\|_2. \tag{13}$$

We equip $\mathcal{X}$ with the Borel σ-algebra $\mathcal{B}(\mathcal{X})$ induced by this metric. Because all trajectories have bounded physical length, each structural coordinate is bounded. We therefore take $\mathcal{X}$ to be a bounded Borel subset of $\mathbb{R}^{12}$ containing $\Phi(\mathcal{Z})$; when compactness is needed, we replace $\mathcal{X}$ by its closure, which leaves the pushforward measures of Appendix A.3 unchanged.

We give the coordinate-level construction of the structural projection

$$\Phi : \mathcal{Z} \to \mathcal{X}, \qquad z \mapsto \boldsymbol{\chi}(z) = [\chi_1(z), \ldots, \chi_{12}(z)]^\top, \tag{14}$$

which maps a variable-length discrete trajectory to a fixed-dimensional structural representation.

Let N denote the number of node references in z, i.e., the node identifiers in `"Generated Node"` and `"Moving to Node"` lines in order of appearance (revisits counted), and let N_{gen} denote the number of explicitly generated nodes. The definitions below assume $N \geq 1$; a trajectory without node references has $\chi_1 = \cdots = \chi_9 = 0$. For the i-th node reference, let id_i denote its sequence identifier and let

$$D_i := |\text{id}_i| \tag{15}$$

be its structural tree depth. Let $C(z, \omega)$ denote the number of exact non-overlapping occurrences of a substring ω in z.

The vector $\boldsymbol{\chi}(z)$ is defined through three groups of statistics. These features summarize the search structure of a trajectory.

Tree-structural and volumetric statistics.

$$\chi_1(z) = \max_{1 \le i \le N} D_i \qquad \text{(maximum depth)}, \tag{16}$$

$$\chi_2(z) = \frac{1}{N} \sum_{i=1}^{N} D_i \qquad \text{(mean depth)}, \tag{17}$$

$$\chi_3(z) = \left(\frac{1}{N} \sum_{i=1}^{N} (D_i - \chi_2)^2 \right)^{1/2} \qquad \text{(depth standard deviation)}, \tag{18}$$

$$\chi_4(z) = N \qquad \text{(search volume)}. \tag{19}$$

Dynamical transition statistics. Let $M := \max(1, N-1)$. We define

$$\chi_5(z) = \frac{1}{M} \sum_{i=1}^{N-1} \mathbb{I}\{D_{i+1} > D_i\} \qquad \text{(deepening rate)}, \tag{20}$$

$$\chi_6(z) = \frac{1}{M} \sum_{i=1}^{N-1} \mathbb{I}\{D_{i+1} < D_i\} \qquad \text{(backtracking rate)}, \tag{21}$$

$$\chi_7(z) = \frac{1}{M} \sum_{i=1}^{N-1} \mathbb{I}\{D_{i+1} = D_i\} \qquad \text{(lateral transition rate)}. \tag{22}$$

Pruning and outcome statistics.

$$\chi_8(z) = C(z, \texttt{"No Solution"}) \qquad \text{(prune count)}, \tag{23}$$

$$\chi_9(z) = \chi_8 / N_{\text{gen}} \qquad \text{(pruning rate; 0 if } N_{\text{gen}} = 0), \tag{24}$$

$$\chi_{10}(z) = \mathbb{I}\{C(z, \texttt{"Goal Reached"}) \ge 1\} \qquad \text{(solution indicator)}, \tag{25}$$

$$\chi_{11}(z) = C(z, \texttt{"*"}) + C(z, \texttt{"/"}) \qquad \text{(multiplicative bias)}, \tag{26}$$

$$\chi_{12}(z) = C(z, \texttt{"+"}) + C(z, \texttt{"-"}) \qquad \text{(additive bias)}. \tag{27}$$

Since $\mathcal{Z}$ is finite and carries the discrete σ-algebra, every map from $\mathcal{Z}$ into $(\mathcal{X}, \mathcal{B}(\mathcal{X}))$ is measurable. In particular, Φ is Borel measurable, and its image $\Phi(\mathcal{Z})$ is finite.

A.2 Theoretical Justification of Structural Sufficiency (ϵ-Relaxation)

Assumption 1 states that the projection $\Phi(z)$ is a sufficient structural proxy for the expected zero-one risk, up to a residual error ϵ. This relaxation is motivated by the deterministic nature of formal combinatorial planning (e.g., the Countdown task). Appendix J finds that $\boldsymbol{\chi}$ alone predicts per-problem success on this task with out-of-fold ROC-AUC 0.93 for both checked models, and that its ROC-AUC exceeds that of token length and of the problem alone in every fold.

Success in algorithmic search tasks is largely determined by the routing strategy of the search. Whether a trajectory reaches a valid solution depends on exploring sufficiently deep computational graphs (captured by maximum depth χ_1), successfully executing recursive recoveries (backtracking rate χ_6), and balancing operator heuristics (additive vs. multiplicative biases χ_{11}, χ_{12}).

By design, our structural feature space $\mathcal{X}$ captures these graph-traversal dynamics. The residual ϵ collects what Φ abstracts away, in particular the token identities (i.e., the specific integer values); it is small when the structure of a problem's reference search, as recorded by Φ, predicts whether the model solves that problem (Appendix J). Given Assumption 1 and a Lipschitz risk function over $\mathcal{X}$, the trajectory-level risk gap is then at most $K_f \, W_1(\nu_S, \nu_T) + 2\epsilon$ (Lemma 1). Thus, the ϵ-relaxation serves as a tractable bridge between discrete sequence generation and continuous measure theory.

A.3 Pushforward Measures and Structural Shift

The structural source and target measures are defined as the pushforwards of μ_S and μ_T under Φ:

$$\nu_S := \Phi_{\#} \mu_S, \qquad \nu_T := \Phi_{\#} \mu_T. \tag{28}$$

Since Φ is Borel measurable, ν_S and ν_T are Borel probability measures on $(\mathcal{X}, \mathcal{B}(\mathcal{X}))$. Moreover, because $\Phi(\mathcal{Z})$ is finite and $\mathcal{X}$ is bounded, both measures have finite first moment; equivalently, $\nu_S, \nu_T \in \mathcal{P}_1(\mathcal{X})$. By definition, for any Borel set $A \in \mathcal{B}(\mathcal{X})$,

$$\nu_S(A) = \mu_S(\Phi^{-1}(A)), \qquad \nu_T(A) = \mu_T(\Phi^{-1}(A)). \tag{29}$$

Because $\mathcal{Z}$ is finite, the pushforward measures admit the explicit representations

$$\nu_S(A) = \sum_{z \in \mathcal{Z}} \mathbb{I}_A(\Phi(z))\, \mu_S(\{z\}), \tag{30}$$

$$\nu_T(A) = \sum_{z \in \mathcal{Z}} \mathbb{I}_A(\Phi(z))\, \mu_T(\{z\}). \tag{31}$$

To quantify the structural discrepancy between source and target reasoning behaviors, we use the Wasserstein-1 distance between their pushforward measures

$$W_1(\nu_S, \nu_T) := \inf_{\pi \in \Pi(\nu_S, \nu_T)} \int_{\mathcal{X} \times \mathcal{X}} d_{\mathcal{X}}(\mathbf{u}, \mathbf{v})\, d\pi(\mathbf{u}, \mathbf{v}), \tag{32}$$

where $\Pi(\nu_S, \nu_T)$ denotes the set of couplings of ν_S and ν_T, i.e., the set of probability measures on $\mathcal{X} \times \mathcal{X}$ whose marginals are ν_S and ν_T [Villani, 2009]. Since $\nu_S, \nu_T \in \mathcal{P}_1(\mathcal{X})$ and $\mathcal{X}$ is bounded, the transportation cost is finite for every $\pi \in \Pi(\nu_S, \nu_T)$. Hence $W_1(\nu_S, \nu_T)$ is well-defined and finite.

A.4 Risk and Structural Risk

Let $\mathcal{F}_{L,m,\mathrm{PE}}$ denote the class of autoregressive Transformer models with layer depth L, hidden dimension m, and positional encoding scheme PE. By the trajectory decomposition defined earlier, each $z \in \mathcal{Z}$ induces an initial problem state, denoted by $x_{\mathrm{init}}^{(z)}$. For a model $f \in \mathcal{F}_{L,m,\mathrm{PE}}$, the generated trajectory obtained from the prompt induced by z is

$$f(x_{\mathrm{init}}^{(z)}) \in \mathcal{Z}_{\mathrm{all}}. \tag{33}$$

Let $S_z : \mathcal{Z}_{\mathrm{all}} \to \mathbb{R}$ be a deterministic scoring function parameterized by the target state of the reference trajectory z, evaluating the formal validity of the generated output. We define the zero-one loss by

$$\ell(f, z) := 1 - \mathbb{I}_{(0,\infty)} \left(S_z \left(f(x_{\mathrm{init}}^{(z)}) \right) \right). \tag{34}$$

For any probability measure $\mu \in \mathcal{P}(\mathcal{Z})$, the expected risk of f under μ is

$$\mathcal{R}_\mu(f) := \mathbb{E}_{z \sim \mu}[\ell(f, z)]. \tag{35}$$

In the main text, we relate the trajectory-level risk $\mathcal{R}_\mu(f)$ to a structural risk functional defined on $\mathcal{X}$ through the pushforward measure $\Phi_{\#}\mu$.

B Proofs of Structural Generalization Results

B.1 Structural Sufficiency Assumption

We recall Assumption 1, which is used throughout the proofs below. It states that there exists an irreducible approximation error $\epsilon > 0$ such that, for each model f, there is a Borel measurable structural risk function $\mathcal{L}_f : \mathcal{X} \to [0, 1]$ satisfying, for every $\mu \in \{\mu_S, \mu_T\}$,

$$\left| \mathcal{R}_\mu(f) - \int_{\mathcal{X}} \mathcal{L}_f(\boldsymbol{\chi})\, d(\Phi_{\#}\mu)(\boldsymbol{\chi}) \right| \leq \epsilon. \tag{36}$$

B.2 Proof of Lemma 1

We now provide a complete proof of Lemma 1, formalizing the sketch given in the main text. For completeness, we restate Lemma 1.

Assume that $\mathcal{L}_f$ is Lipschitz continuous on $(\mathcal{X}, d_{\mathcal{X}})$ with minimal Lipschitz constant

$$K_f := \sup_{\boldsymbol{u} \neq \boldsymbol{v} \in \mathcal{X}} \frac{|\mathcal{L}_f(\boldsymbol{u}) - \mathcal{L}_f(\boldsymbol{v})|}{d_{\mathcal{X}}(\boldsymbol{u}, \boldsymbol{v})}. \tag{37}$$

Then the generalization error gap satisfies

$$|\mathcal{R}_{\mu_T}(f) - \mathcal{R}_{\mu_S}(f)| \leq K_f \, W_1(\nu_S, \nu_T) + 2\epsilon. \tag{38}$$

Proof. Let

$$\Delta\mathcal{R} := |\mathcal{R}_{\mu_T}(f) - \mathcal{R}_{\mu_S}(f)| \,. \tag{39}$$

Adding and subtracting the structural expectations under ν_T and ν_S gives

$$\begin{aligned} \Delta\mathcal{R} = \Bigg| & \mathcal{R}_{\mu_T}(f) - \int_{\mathcal{X}} \mathcal{L}_f(\boldsymbol{\chi}) \, d\nu_T(\boldsymbol{\chi}) + \int_{\mathcal{X}} \mathcal{L}_f(\boldsymbol{\chi}) \, d\nu_T(\boldsymbol{\chi}) \\ & - \int_{\mathcal{X}} \mathcal{L}_f(\boldsymbol{\chi}) \, d\nu_S(\boldsymbol{\chi}) + \int_{\mathcal{X}} \mathcal{L}_f(\boldsymbol{\chi}) \, d\nu_S(\boldsymbol{\chi}) - \mathcal{R}_{\mu_S}(f) \Bigg|. \end{aligned} \tag{40}$$

Applying the triangle inequality yields

$$\begin{aligned} \Delta\mathcal{R} \leq & \left| \mathcal{R}_{\mu_T}(f) - \int_{\mathcal{X}} \mathcal{L}_f(\boldsymbol{\chi}) \, d\nu_T(\boldsymbol{\chi}) \right| \\ & + \left| \int_{\mathcal{X}} \mathcal{L}_f(\boldsymbol{\chi}) \, d\nu_T(\boldsymbol{\chi}) - \int_{\mathcal{X}} \mathcal{L}_f(\boldsymbol{\chi}) \, d\nu_S(\boldsymbol{\chi}) \right| \\ & + \left| \int_{\mathcal{X}} \mathcal{L}_f(\boldsymbol{\chi}) \, d\nu_S(\boldsymbol{\chi}) - \mathcal{R}_{\mu_S}(f) \right|. \end{aligned} \tag{41}$$

By Assumption 1,

$$\Delta\mathcal{R} \leq \left| \int_{\mathcal{X}} \mathcal{L}_f(\boldsymbol{\chi}) \, d\nu_T(\boldsymbol{\chi}) - \int_{\mathcal{X}} \mathcal{L}_f(\boldsymbol{\chi}) \, d\nu_S(\boldsymbol{\chi}) \right| + 2\epsilon. \tag{42}$$

If $K_f = 0$, then $\mathcal{L}_f$ is constant on $\mathcal{X}$, so the integral difference vanishes and the claim follows immediately.

Assume now that $K_f > 0$, and define

$$g(\boldsymbol{\chi}) := \frac{\mathcal{L}_f(\boldsymbol{\chi})}{K_f}. \tag{43}$$

Then g is 1-Lipschitz. Since $\nu_S, \nu_T \in \mathcal{P}_1(\mathcal{X})$, the Kantorovich–Rubinstein duality for W_1 applies on $(\mathcal{X}, d_{\mathcal{X}})$ [Villani, 2009, Weed and Bach, 2019]. Therefore,

$$\left| \int_{\mathcal{X}} g(\boldsymbol{\chi}) \, d\nu_T(\boldsymbol{\chi}) - \int_{\mathcal{X}} g(\boldsymbol{\chi}) \, d\nu_S(\boldsymbol{\chi}) \right| \leq W_1(\nu_S, \nu_T). \tag{44}$$

Substituting $g = \mathcal{L}_f / K_f$ gives

$$\left| \int_{\mathcal{X}} \mathcal{L}_f(\boldsymbol{\chi}) \, d\nu_T(\boldsymbol{\chi}) - \int_{\mathcal{X}} \mathcal{L}_f(\boldsymbol{\chi}) \, d\nu_S(\boldsymbol{\chi}) \right| \leq K_f \, W_1(\nu_S, \nu_T). \tag{45}$$

Combining this bound with the previous inequality yields

$$\Delta\mathcal{R} \leq K_f \, W_1(\nu_S, \nu_T) + 2\epsilon, \tag{46}$$

which proves the claim. □

C Proofs of Positional Encoding Results

C.1 Attention Kernels and the Translation Operator

We first restate the attention kernels for APE and RoPE. Let $\mathbf{x}_i, \mathbf{x}_j \in \mathbb{R}^d$ denote hidden state vectors at absolute sequence indices i and j, and let $\mathbf{W}_Q, \mathbf{W}_K \in \mathbb{R}^{d\times d}$ denote the learned query and key projection matrices.

For Absolute Positional Encoding (APE), let $\mathbf{p}_i, \mathbf{p}_j \in \mathbb{R}^d$ denote the absolute position embeddings. The pre-softmax attention score is

$$A_{i,j}^{\mathrm{APE}} = (\mathbf{x}_i + \mathbf{p}_i)^\top \mathbf{W}_Q^\top \mathbf{W}_K (\mathbf{x}_j + \mathbf{p}_j). \tag{47}$$

For Rotary Position Embedding (RoPE) [Su et al., 2024], let $\mathbf{R}_\Theta^t \in \mathbb{R}^{d\times d}$ denote the orthogonal block-diagonal rotation matrix associated with position t. Using $(\mathbf{R}_\Theta^i)^\top = (\mathbf{R}_\Theta^i)^{-1} = \mathbf{R}_\Theta^{-i}$, the pre-softmax attention score is

$$\begin{aligned} A_{i,j}^{\mathrm{RoPE}} &= (\mathbf{R}_\Theta^i \mathbf{W}_Q \mathbf{x}_i)^\top (\mathbf{R}_\Theta^j \mathbf{W}_K \mathbf{x}_j) && (48)\\ &= \mathbf{x}_i^\top \mathbf{W}_Q^\top (\mathbf{R}_\Theta^i)^\top \mathbf{R}_\Theta^j \mathbf{W}_K \mathbf{x}_j && (49)\\ &= \mathbf{x}_i^\top \mathbf{W}_Q^\top \mathbf{R}_\Theta^{j-i} \mathbf{W}_K \mathbf{x}_j. && (50) \end{aligned}$$

Operational Bounds on Structural Perturbation δ. Let $z \in \mathcal{Z}$ be written as $z = x_{\mathrm{init}}^{(z)} \oplus y_{\mathrm{search}}$. For $k \in \mathbb{N}^+$, the translation operator

$$\mathcal{T}_k : \mathcal{Z} \to \mathcal{Z}$$

inserts an isolated, structurally inert node (e.g., an identity operation such as `x=x`) of exactly k tokens immediately before a local reasoning sub-trajectory $\omega \subset y_{\mathrm{search}}$.

Under the projection Φ, this insertion increments the absolute volume coordinate by one: $\chi_4' = \chi_4 + 1$. Because the inserted node does not alter the maximum tree depth ($\Delta\chi_1 = 0$) and its contribution to the global transition rates and depth variance decays at a rate of $\mathcal{O}(1/N)$, the remaining structural perturbations are negligible for long trajectories. Hence, the exact Euclidean structural distance evaluates to:

$$d_{\mathcal{X}}(\boldsymbol{\chi}, \boldsymbol{\chi}') = \left(1^2 + \sum_{m\neq 4} (\Delta\chi_m)^2\right)^{1/2} =: \delta. \tag{51}$$

As the trajectory length N grows, $\sum_{m\neq 4}(\Delta\chi_m)^2 \to 0$, which bounds the perturbation distance around unity: $1 \le \delta \le 1 + \mathcal{O}(1/N) =: C_\delta$. Concurrently, every token in the subsequent sub-trajectory ω undergoes an absolute index shift of $+k$.

C.2 Proof of Lemma 2

For completeness, we restate Lemma 2.

Assume the network parameters (projection matrices $\mathbf{W}_Q, \mathbf{W}_K$ and the absolute position embeddings $\mathbf{p}_t$) are generically parameterized (i.e., avoiding exact algebraic degeneracies), and that

$$\mathbf{W}_Q^\top \mathbf{W}_K \neq \mathbf{0}. \tag{52}$$

Then, under the translation operator $\mathcal{T}_k$ with $k > 0$, the local attention kernel of an APE model differs almost surely:

$$\Delta A^{\mathrm{APE}} = |\tilde{A}_{a,b}^{\mathrm{APE}} - A_{a,b}^{\mathrm{APE}}| > 0 \qquad \text{a.s.} \tag{53}$$

Proof. Let $\mathbf{x}_a, \mathbf{x}_b \in \omega$ be tokens originally appearing at indices a and b. After applying $\mathcal{T}_k$, these indices become $a + k$ and $b + k$. The resulting attention difference is

$$\begin{aligned} \Delta A^{\mathrm{APE}} &:= \left|\widetilde{A}_{a,b}^{\mathrm{APE}} - A_{a,b}^{\mathrm{APE}}\right| && (54)\\ &= \left|(\mathbf{x}_a + \mathbf{p}_{a+k})^\top \mathbf{W}_Q^\top \mathbf{W}_K (\mathbf{x}_b + \mathbf{p}_{b+k}) - (\mathbf{x}_a + \mathbf{p}_a)^\top \mathbf{W}_Q^\top \mathbf{W}_K (\mathbf{x}_b + \mathbf{p}_b)\right|. && (55) \end{aligned}$$

Expanding both bilinear forms and canceling the shared term $\mathbf{x}_a^\top \mathbf{W}_Q^\top \mathbf{W}_K \mathbf{x}_b$ yields

$$\Delta A^{\mathrm{APE}} = \Big| \mathbf{x}_a^\top \mathbf{W}_Q^\top \mathbf{W}_K (\mathbf{p}_{b+k} - \mathbf{p}_b) + (\mathbf{p}_{a+k} - \mathbf{p}_a)^\top \mathbf{W}_Q^\top \mathbf{W}_K \mathbf{x}_b + \mathbf{p}_{a+k}^\top \mathbf{W}_Q^\top \mathbf{W}_K \mathbf{p}_{b+k} - \mathbf{p}_a^\top \mathbf{W}_Q^\top \mathbf{W}_K \mathbf{p}_b \Big|. \tag{56}$$

Introduce $\mathbf{u} := \mathbf{p}_{a+k}$ and $\mathbf{v} := \mathbf{p}_{b+k}$, and define

$$S(\mathbf{u}, \mathbf{v}) := \mathbf{u}^\top \mathbf{W}_Q^\top \mathbf{W}_K \mathbf{v} + \mathbf{x}_a^\top \mathbf{W}_Q^\top \mathbf{W}_K \mathbf{v} + \mathbf{u}^\top \mathbf{W}_Q^\top \mathbf{W}_K \mathbf{x}_b - \Big(\mathbf{x}_a^\top \mathbf{W}_Q^\top \mathbf{W}_K \mathbf{p}_b + \mathbf{p}_a^\top \mathbf{W}_Q^\top \mathbf{W}_K \mathbf{x}_b + \mathbf{p}_a^\top \mathbf{W}_Q^\top \mathbf{W}_K \mathbf{p}_b \Big). \tag{57}$$

Then perfect cancellation would require $S(\mathbf{u}, \mathbf{v}) = 0$.

The leading term of S is the bilinear form

$$\mathbf{u}^\top \mathbf{W}_Q^\top \mathbf{W}_K \mathbf{v} = \sum_{i=1}^{d} \sum_{j=1}^{d} u_i (\mathbf{W}_Q^\top \mathbf{W}_K)_{ij} v_j. \tag{58}$$

Since $\mathbf{W}_Q^\top \mathbf{W}_K \neq \mathbf{0}$, this is a nontrivial polynomial in $(\mathbf{u}, \mathbf{v}) \in \mathbb{R}^{2d}$. Therefore by the properties of real algebraic geometry [Mityagin, 2020], the zero set

$$\{(\mathbf{u}, \mathbf{v}) \in \mathbb{R}^{2d} : S(\mathbf{u}, \mathbf{v}) = 0\} \tag{59}$$

is a proper algebraic variety and has Lebesgue measure zero.

Satisfying this exact cancellation for arbitrary sequence shifts k requires the positional embeddings to exhibit strict periodic symmetries or precise null-space alignments. Under generic parameterization of fully trained dense neural networks, the parameters avoid this specific measure-zero algebraic variety almost everywhere. Hence

$$\Delta A^{\mathrm{APE}} > 0 \qquad \text{almost surely.} \tag{60}$$

□

C.3 RoPE Equivariance under Sequence Translation

We provide the RoPE invariance argument used in the proof of Theorem 1. Under the translation operator $\mathcal{T}_k$, the local attention kernel of RoPE is invariant: $\Delta A^{\mathrm{RoPE}} = 0$.

Proof. Let $\mathbf{x}_a, \mathbf{x}_b$ be translated from indices a, b to $a + k, b + k$. The translated RoPE kernel is

$$\widetilde{A}_{a,b}^{\mathrm{RoPE}} = \mathbf{x}_a^\top \mathbf{W}_Q^\top \mathbf{R}_\Theta^{(b+k)-(a+k)} \mathbf{W}_K \mathbf{x}_b \tag{61}$$

$$= \mathbf{x}_a^\top \mathbf{W}_Q^\top \mathbf{R}_\Theta^{b-a} \mathbf{W}_K \mathbf{x}_b \tag{62}$$

$$= A_{a,b}^{\mathrm{RoPE}}. \tag{63}$$

Therefore

$$\Delta A^{\mathrm{RoPE}} = \left| \widetilde{A}_{a,b}^{\mathrm{RoPE}} - A_{a,b}^{\mathrm{RoPE}} \right| = 0. \tag{64}$$

□

C.4 Details of Assumption 2

We expand Assumption 2, which is used in the proof of Theorem 1.

(i) Global smoothness. The underlying reasoning task admits an intrinsic structural smoothness scale satisfying $K_{\mathrm{base}} = \mathcal{O}(1/T_{\max})$. Because RoPE preserves relative position information exactly under sequence translations, it inherits this smoothness scale: $K_{f_{\mathrm{RoPE}}} \leq \mathcal{O}(1/T_{\max})$.

(ii) Autoregressive sensitivity and fragile logit margin. Let $f_{\mathrm{APE}} \in \mathcal{F}_{L,m,\mathrm{APE}}$. In formal combinatorial planning, generating a sequence operates as a deterministic Markov decision process. We assume that structurally critical routing steps (e.g., recursive backtracking steps in deep subtrees)

exhibit a fragile logit margin. Let $\Delta_{\mathrm{margin}} > 0$ denote the minimal pre-softmax logit separation required between the Bayes-optimal routing token $a^\star$ and competing incorrect tokens. For out-of-distribution reasoning trajectories with deep hierarchical structures, we assume this margin shrinks ($\Delta_{\mathrm{margin}} \to 0^+$). Consequently, once the margin falls below the disruption ($\Delta_{\mathrm{margin}} < \Delta A$, with $\Delta A > 0$ induced by APE translations; Lemma 2), the deterministic greedy decoding process selects an incorrect subtree. This precludes reaching the target state, bounding the expected structural risk variation below by the deterministic separation margin:

$$|\mathcal{L}_{f_{\mathrm{APE}}}(\boldsymbol{\chi}') - \mathcal{L}_{f_{\mathrm{APE}}}(\boldsymbol{\chi})| \geq \gamma, \qquad \gamma := 1 - \frac{1}{|\mathcal{V}|} > 0. \tag{65}$$

Routing separation under deterministic decoding. Let $a^\star(z)$ denote the Bayes-optimal routing action at the critical transition of trajectory z, and let

$$a_f(z) := \arg\max_{a \in \mathcal{V}} p_f(a \mid h_z)$$

denote the action selected by the model under greedy decoding, where h_z is the prefix of the model's own greedy trace for the problem $x_{\mathrm{init}}^{(z)}$ up to that transition. Assumption 2 postulates that, at a structurally critical transition, the APE-induced coordinate perturbation changes the selected routing action away from the Bayes-optimal action,

$$a_f(z') \neq a^\star(z'),$$

on problems z' whose reference searches map near $\boldsymbol{\chi}' = \Phi(\mathcal{T}_k z)$ (see the next paragraph). This routing deviation sends the autoregressive process into an incorrect subtree and prevents the generated trajectory for z' from reaching the target state. Here an incorrect subtree means a routing error after which the model's own trace reports no valid path to the target within the 512-token generation budget. Because the scorer gives no partial credit (it checks the path that the trace reports to the target; Appendix K), the zero-one loss on z' is then 1. Hence the induced structural risk changes by a constant amount independent of $T_{\max}$:

$$|\mathcal{L}_{f_{\mathrm{APE}}}(\boldsymbol{\chi}') - \mathcal{L}_{f_{\mathrm{APE}}}(\boldsymbol{\chi})| \geq \gamma, \qquad \gamma := 1 - \frac{1}{|\mathcal{V}|} > 0. \tag{66}$$

The constant $1 - 1/|\mathcal{V}|$ is the error rate of uniform guessing over $\mathcal{V}$. Under exact-match evaluation the jump can be as large as 1; for Theorem 1 it suffices that $\gamma > 0$ is independent of $T_{\max}$.

What Assumption 2 constrains. The loss $\ell(f, z)$ depends on z only through the problem $x_{\mathrm{init}}^{(z)}$ and its target (Appendix A.4), while $\mathcal{T}_k$ acts on y_{search}, so $\ell(f, \mathcal{T}_k z) = \ell(f, z)$. Assumption 2 therefore constrains how the structural risk function $\mathcal{L}_f$ varies between the neighboring points $\boldsymbol{\chi} = \Phi(z)$ and $\boldsymbol{\chi}' = \Phi(\mathcal{T}_k z)$ of $\mathcal{X}$. On other problems z' whose reference searches map near $\boldsymbol{\chi}'$, an APE model places a familiar sub-search of its own trace at a shifted absolute position and thus undergoes the kernel change of Lemma 2; under Assumption 2(ii), this change yields the jump γ.

C.5 Proof of Theorem 1

For completeness, we restate Theorem 1, which holds under Assumption 2. For sufficiently large $T_{\max}$, let $f_{\mathrm{APE}} \in \mathcal{F}_{L,m,\mathrm{APE}}$ and $f_{\mathrm{RoPE}} \in \mathcal{F}_{L,m,\mathrm{RoPE}}$ be two matched autoregressive Transformer models that differ only in their positional encoding scheme. Then

$$K_{f_{\mathrm{RoPE}}} = \mathcal{O}(1/T_{\max}), \qquad K_{f_{\mathrm{APE}}} = \Omega(1), \tag{67}$$

and hence $K_{f_{\mathrm{RoPE}}} \ll K_{f_{\mathrm{APE}}}$.

Proof. By definition,

$$K_{f_{\mathrm{APE}}} = \sup_{\boldsymbol{u} \neq \boldsymbol{v} \in \mathcal{X}} \frac{|\mathcal{L}_{f_{\mathrm{APE}}}(\boldsymbol{u}) - \mathcal{L}_{f_{\mathrm{APE}}}(\boldsymbol{v})|}{d_{\mathcal{X}}(\boldsymbol{u}, \boldsymbol{v})}. \tag{68}$$

Let $z \in \mathcal{Z}$ be a trajectory whose translated sub-trajectory under $\mathcal{T}_k$ contains a critical routing step in the sense of Assumption 2(ii), and let

$$\boldsymbol{\chi} = \Phi(z), \qquad \boldsymbol{\chi}' = \Phi(\mathcal{T}_k(z)). \tag{69}$$

By Definition 2, the operator $\mathcal{T}_k$ inserts an inert node immediately before a local reasoning subtrajectory. By Lemma 2, this induces a nonzero local APE attention perturbation almost surely. Assumption 2(ii), read as in the paragraph *What Assumption 2 constrains* of Appendix C.4, then yields

$$|\mathcal{L}_{f_{\mathrm{APE}}}(\boldsymbol{\chi}') - \mathcal{L}_{f_{\mathrm{APE}}}(\boldsymbol{\chi})| \geq \gamma. \tag{70}$$

Hence

$$K_{f_{\mathrm{APE}}} \geq \frac{|\mathcal{L}_{f_{\mathrm{APE}}}(\boldsymbol{\chi}') - \mathcal{L}_{f_{\mathrm{APE}}}(\boldsymbol{\chi})|}{d_{\mathcal{X}}(\boldsymbol{\chi}, \boldsymbol{\chi}')} \geq \frac{\gamma}{\delta} \geq \frac{\gamma}{C_\delta} = \Omega(1). \tag{71}$$

Assumption 2(i) gives

$$K_{f_{\mathrm{RoPE}}} \leq \mathcal{O}(1/T_{\max}). \tag{72}$$

Therefore, for sufficiently large $T_{\max}$,

$$K_{f_{\mathrm{RoPE}}} \ll K_{f_{\mathrm{APE}}}. \tag{73}$$

□

D Proofs of Capacity Bounds

D.1 Structural Capacity Analysis

Definition 3 (Bayes-Optimal Structural Risk)**.** *The Bayes-optimal structural risk function is defined by*

$$\mathcal{L}^*(\boldsymbol{\chi}) := \inf_h \mathbb{E}_{z\sim\mu_T}[\ell(h, z) \mid \Phi(z) = \boldsymbol{\chi}], \qquad \boldsymbol{\chi} \in \mathcal{X}. \tag{74}$$

Definition 4 (Barron Space)**.** *Let $\mathcal{X} \subset \mathbb{R}^{12}$ be compact. The Barron space $\mathcal{H}_{\mathrm{Barron}}(\mathcal{X})$ consists of continuous functions $g : \mathcal{X} \to \mathbb{R}$ whose Fourier transform $\hat{g}(\boldsymbol{\omega})$ satisfies*

$$\int_{\mathbb{R}^{12}} \|\boldsymbol{\omega}\|_1 \, |\hat{g}(\boldsymbol{\omega})| \, d\boldsymbol{\omega} < \infty. \tag{75}$$

The infimum of this quantity defines the Barron norm $\|g\|_{\mathcal{B}}$ [Barron, 1993].

Assumption 3 (Tsybakov Margin Condition)**.** *There exist constants $C > 0$ and $\alpha \geq 0$ such that for the conditional label probability $\eta(\boldsymbol{\chi}) := \mathbb{P}_{z\sim\mu_T}(\ell(f, z) = 1 \mid \Phi(z) = \boldsymbol{\chi})$,*

$$\nu_T\big(|\eta(\boldsymbol{\chi}) - 1/2| \leq t\big) \leq Ct^\alpha, \quad \forall t > 0.$$

Assumption 3 is a standard condition for relating the $L_1(\nu_T)$ approximation error of a structural risk function to the zero-one risk [Tsybakov, 2004]. The proof of Lemma 4 (Appendix D.1.1) measures this error against the pointwise Bayes risk $\mathcal{L}^*$ (Definition 3), which gives the relation directly, with constant $\frac{1}{2}$ and residual $\frac{\epsilon}{2}$, so this proof rests on Assumptions 1 and 4.

Assumption 4 (Structural Capacity Mapping and Approximation Limit)**.** *We assume the following.*
(i) Barron regularity of the target risk. *The Bayes-optimal structural risk belongs to the Barron space:*

$$\mathcal{L}^* \in \mathcal{H}_{\mathrm{Barron}}(\mathcal{X}), \qquad 0 < \|\mathcal{L}^*\|_{\mathcal{B}} < \infty. \tag{76}$$

(ii) Finite-width structural hypothesis class. *For each autoregressive Transformer $f \in \mathcal{F}_{L,m,\mathrm{PE}}$, the induced structural risk function $\mathcal{L}_f$ belongs to a width-constrained hypothesis class*

$$\mathcal{L}_f \in \mathcal{H}_m \subset \mathcal{H}_{\mathrm{Barron}}(\mathcal{X}), \tag{77}$$

where $\mathcal{H}_m$ denotes a structural approximation class whose capacity is controlled by the internal representation width m.

(iii) Width-limited approximation lower bound. *There exists a constant $C_{\mathcal{B}} > 0$, independent of m, such that*

$$\inf_{g\in\mathcal{H}_m} \int_{\mathcal{X}} |g(\boldsymbol{\chi}) - \mathcal{L}^*(\boldsymbol{\chi})| \, d\nu_T(\boldsymbol{\chi}) \geq \frac{C_{\mathcal{B}}}{\sqrt{m}}. \tag{78}$$

This scaling mirrors the $\mathcal{O}(m^{-1/2})$ upper rate for Barron-type function classes [Barron, 1993].

Assumption 4 formalizes the idea that projecting discrete trajectories into $\mathcal{X}$ induces a regression problem whose best attainable predictor cannot be approximated arbitrarily well at fixed width.

D.1.1 Proof of Lemma 4

Proof. By Assumption 1, the true expected zero-one risk is bounded by the structural risk:

$$\mathcal{R}_{\mu_T}(f) \geq \int_{\mathcal{X}} \mathcal{L}_f(\boldsymbol{\chi})\, d\nu_T(\boldsymbol{\chi}) - \epsilon. \tag{79}$$

Since $\mathcal{L}^*$ is the pointwise Bayes risk, $\int_{\mathcal{X}} \mathcal{L}^*\, d\nu_T \leq \mathcal{R}_{\mu_T}(f)$; with the bound above and the inequality $|\mathcal{L}_f - \mathcal{L}^*| \leq \mathcal{L}_f + \mathcal{L}^*$, this gives $\mathcal{R}_{\mu_T}(f) \geq \frac{1}{2}\int_{\mathcal{X}} |\mathcal{L}_f - \mathcal{L}^*|\, d\nu_T - \frac{\epsilon}{2}$.

Since $\mathcal{L}_f \in \mathcal{H}_m$, Assumption 4(iii) (the minimax-hardness of $\mathcal{L}^*$) implies that recovering $\mathcal{L}^*$ incurs an absolute L_1 error of at least $C_{\mathcal{B}}\, m^{-1/2}$. Thus:

$$\int_{\mathcal{X}} |\mathcal{L}_f(\boldsymbol{\chi}) - \mathcal{L}^*(\boldsymbol{\chi})|\; d\nu_T(\boldsymbol{\chi}) \geq \frac{C_{\mathcal{B}}}{\sqrt{m}}. \tag{80}$$

Defining $C_{\text{width}} := C_{\mathcal{B}}/2$ and substituting this back yields $\mathcal{R}_{\mu_T}(f) \geq \frac{C_{\text{width}}}{\sqrt{m}} - \epsilon$. □

D.2 Depth Bottleneck via Dyck-Type Reduction

We now give a more explicit formalization of the depth bottleneck underlying sequential backtracking. The central idea is that nested backtracking induces a Dyck-type dependency whose effective computational depth scales with the structural depth coordinate χ_1.

D.2.1 DFS-to-Dyck Encoding

We give the detailed construction underlying Lemma 3. Let k denote the maximum branching factor of the search tree. We define the search transition alphabet

$$\Sigma_{\text{search}} := \{\delta_{\text{down}}^{(1)}, \ldots, \delta_{\text{down}}^{(k)}\} \cup \{\delta_{\text{up}}^{(1)}, \ldots, \delta_{\text{up}}^{(k)}\} \cup \{\omega_{\text{eval}}\}, \tag{81}$$

where $\delta_{\text{down}}^{(c)}$ denotes expansion to the c-th child, $\delta_{\text{up}}^{(c)}$ denotes backtracking from the c-th child, and ω_{eval} denotes terminal evaluation.

Let $\Sigma_{\text{Dyck}} = \{(_1,)_1, \ldots, (_k,)_k\}$, and let $\mathcal{D}_k \subset \Sigma_{\text{Dyck}}^*$ denote the Dyck-k language [Strobl et al., 2024]. Define the map

$$\psi : \Sigma_{\text{search}}^* \to \Sigma_{\text{Dyck}}^* \tag{82}$$

pointwise by

$$\psi(\delta_{\text{down}}^{(c)}) = (_c, \qquad \psi(\delta_{\text{up}}^{(c)}) =)_c, \qquad \psi(\omega_{\text{eval}}) = \varepsilon. \tag{83}$$

Since ψ preserves concatenation, i.e., $\psi(x \oplus y) = \psi(x) \oplus \psi(y)$ for all $x, y \in \Sigma_{\text{search}}^*$, it is a monoid homomorphism.

When a valid trajectory z is read as the walk on the search tree through its successively visited nodes, with each transition expanded into parent-child moves (for DFS these are exactly the push and pop moves of backtracking), its image $\psi(z)$ records this nested push-pop structure as a prefix of a balanced-parenthesis string, which is balanced if and only if the walk returns to the root. In particular, the maximum nesting depth of $\psi(z)$ equals the maximum depth reached, i.e., $\chi_1 - 1$ under the identifier convention of Appendix A.1.

D.2.2 Complexity-Theoretic Interpretation and Finite Precision

Consider an autoregressive step t at which the next token corresponds to a backtracking move $v_{t+1} = \delta_{\text{up}}^{(c)}$. To emit this token correctly, the model must identify the most recent unmatched opening transition associated with the same branch type c. Under the encoding above, this is equivalent to resolving a nested Dyck-type dependency over the serialized search history.

Parsing Dyck-language dependencies relies on maintaining a stack counter within the continuous hidden representations. For a Transformer utilizing bounded numerical precision with p bits (e.g., $p = 16$ for bfloat16), the capacity to reliably encode distinguishable nested states is information-theoretically constrained by $\mathcal{O}(2^p)$. As shown by Hahn [2020], self-attention cannot model hierarchical languages unless the number of layers or heads grows with the input length; we therefore treat the evaluation of a nested structure as requiring sequential reductions across the layers.

We formalize this depth requirement by letting the maximum resolvable tree depth χ_1 scale linearly with the physical layer depth L (Assumption 5 below). The truncation constant $\alpha_{\mathrm{circ}} := c^{-1}$ defined there, which may depend on the task and on the numerical precision p, sets the maximum parsable tree depth per physical layer.

Assumption 5 (Implicit State-Transition Complexity via Dyck Structure)**.** *There exists a precision-dependent constant $c > 0$ such that resolving a backtracking transition of structural depth χ_1 requires implicit computational depth at least*

$$D_{\mathrm{implicit}} \geq c\,\chi_1. \tag{84}$$

Equivalently, defining $\alpha_{\mathrm{circ}} := c^{-1}$, trajectories satisfying $\chi_1 > \alpha_{\mathrm{circ}} L$ lie strictly beyond the implicit computational depth available within a single autoregressive forward pass. On these trajectories every $f \in \mathcal{F}_{L,m,\mathrm{PE}}$ has structural risk at least that of uniform guessing over $\mathcal{V}$: $\mathcal{L}_f(\boldsymbol{\chi}) \geq 1 - 1/|\mathcal{V}|$ whenever $\chi_1 > \alpha_{\mathrm{circ}} L$.

D.2.3 Failure Beyond the Implicit Depth Threshold

Define the severed region

$$\mathcal{X}_{\mathrm{severed}} := \{\boldsymbol{\chi} \in \mathcal{X} : \chi_1 > \alpha_{\mathrm{circ}} L\}. \tag{85}$$

For trajectories in $\mathcal{X}_{\mathrm{severed}}$, the model lacks sufficient implicit depth to resolve the required backtracking dependency. In this regime, the relevant routing decision can no longer be reliably distinguished from competing valid outputs, and Assumption 5 lower-bounds the resulting structural risk by the uniform-guessing baseline

$$C_{\mathrm{depth}} := 1 - \frac{1}{|\mathcal{V}|}. \tag{86}$$

This motivates the truncation indicator

$$P_{\mathrm{implicit}}(\boldsymbol{\chi}; L, \alpha_{\mathrm{circ}}) := \mathbb{I}_{(\alpha_{\mathrm{circ}} L, \infty)}(\chi_1), \tag{87}$$

which records whether a structural state lies beyond the implicit depth threshold.

D.2.4 Proof of Lemma 5

For completeness, we restate Lemma 5, which holds under Assumptions 1 and 5. Let $|\mathcal{V}| \geq 2$, and define $C_{\mathrm{depth}} := 1 - \frac{1}{|\mathcal{V}|} \in (0, 1)$. Then for any autoregressive Transformer hypothesis $f \in \mathcal{F}_{L,m,\mathrm{PE}}$,

$$\mathcal{R}_{\mu_T}(f) \geq C_{\mathrm{depth}} \int_{\mathcal{X}} P_{\mathrm{implicit}}(\boldsymbol{\chi}; L, \alpha_{\mathrm{circ}})\, d\nu_T(\boldsymbol{\chi}) - \epsilon. \tag{88}$$

Proof. For $\boldsymbol{\chi} \in \mathcal{X}_{\mathrm{severed}}$, the required implicit routing depth exceeds the available physical depth L, and Assumption 5 lower-bounds the structural risk on this region by

$$\mathcal{L}_f(\boldsymbol{\chi}) \geq C_{\mathrm{depth}}. \tag{89}$$

It follows that

$$\int_{\mathcal{X}} \mathcal{L}_f(\boldsymbol{\chi})\, d\nu_T(\boldsymbol{\chi}) = \int_{\mathcal{X}_{\mathrm{severed}}} \mathcal{L}_f(\boldsymbol{\chi})\, d\nu_T(\boldsymbol{\chi}) + \int_{\mathcal{X} \setminus \mathcal{X}_{\mathrm{severed}}} \mathcal{L}_f(\boldsymbol{\chi})\, d\nu_T(\boldsymbol{\chi}) \tag{90}$$

$$\geq \int_{\mathcal{X}_{\mathrm{severed}}} \mathcal{L}_f(\boldsymbol{\chi})\, d\nu_T(\boldsymbol{\chi}) \tag{91}$$

$$\geq C_{\mathrm{depth}}\, \nu_T(\mathcal{X}_{\mathrm{severed}}). \tag{92}$$

Since

$$\nu_T(\mathcal{X}_{\mathrm{severed}}) = \int_{\mathcal{X}} P_{\mathrm{implicit}}(\boldsymbol{\chi}; L, \alpha_{\mathrm{circ}})\, d\nu_T(\boldsymbol{\chi}), \tag{93}$$

we obtain

$$\int_{\mathcal{X}} \mathcal{L}_f(\boldsymbol{\chi})\, d\nu_T(\boldsymbol{\chi}) \geq C_{\mathrm{depth}} \int_{\mathcal{X}} P_{\mathrm{implicit}}(\boldsymbol{\chi}; L, \alpha_{\mathrm{circ}})\, d\nu_T(\boldsymbol{\chi}). \tag{94}$$

Finally, Assumption 1 gives

$$\mathcal{R}_{\mu_T}(f) \geq \int_{\mathcal{X}} \mathcal{L}_f(\boldsymbol{\chi})\, d\nu_T(\boldsymbol{\chi}) - \epsilon, \tag{95}$$

which yields the claim. □

D.2.5 Architecture-Invariance of the Physical Depth Limit

Lemma 6 (Architecture-Invariance of the Physical Depth Limit)**.** *The scalar α_{circ} is determined by the implicit computational requirement of the backtracking problem and is therefore independent of the positional encoding scheme. In particular, $\alpha_{\mathrm{APE}} = \alpha_{\mathrm{RoPE}} = \alpha_{\mathrm{circ}}$.*

Proof. A single autoregressive forward pass applies exactly L sequential layers, regardless of whether the positional encoding is APE or RoPE. RoPE preserves relative position information (Appendix C.3) but adds no sequential computational steps beyond the available physical depth. Therefore the same implicit depth threshold governs both positional-encoding choices. □

E Proof of the Main Theorem and Corollaries

E.1 Proof of Theorem 2

For completeness, we restate Theorem 2, which holds under Assumptions 1, 4 and 5 and the Lipschitz continuity of $\mathcal{L}_f$ assumed in Lemma 1. For any autoregressive Transformer hypothesis $f \in \mathcal{F}_{L,m,\mathrm{PE}}$, the target risk satisfies

$$\mathcal{R}_{\mu_T}(f) \geq \max \left\{ \begin{array}{c} \dfrac{C_{\mathrm{width}}}{\sqrt{m}} - \epsilon,\ C_{\mathrm{depth}} \displaystyle\int_{\mathcal{X}} P_{\mathrm{implicit}}(\boldsymbol{\chi}; L, \alpha_{\mathrm{circ}})\, d\nu_T(\boldsymbol{\chi}) - \epsilon, \\ \mathcal{R}_{\mu_S}(f) - K_{f,\mathrm{PE}} W_1(\nu_S, \nu_T) - 2\epsilon \end{array} \right\}. \tag{96}$$

Moreover, $\mathcal{R}_{\mu_T}(f) \leq \mathcal{R}_{\mu_S}(f) + K_{f,\mathrm{PE}} W_1(\nu_S, \nu_T) + 2\epsilon$.

Proof. Lemma 4 gives $\mathcal{R}_{\mu_T}(f) \geq \frac{C_{\mathrm{width}}}{\sqrt{m}} - \epsilon$. Lemma 5 gives $\mathcal{R}_{\mu_T}(f) \geq C_{\mathrm{depth}} \int_{\mathcal{X}} P_{\mathrm{implicit}}(\boldsymbol{\chi}; L, \alpha_{\mathrm{circ}})\, d\nu_T(\boldsymbol{\chi}) - \epsilon$. Finally, Lemma 1 implies $|\mathcal{R}_{\mu_T}(f) - \mathcal{R}_{\mu_S}(f)| \leq K_{f,\mathrm{PE}} W_1(\nu_S, \nu_T) + 2\epsilon$, and therefore $\mathcal{R}_{\mu_T}(f) \geq \mathcal{R}_{\mu_S}(f) - K_{f,\mathrm{PE}} W_1(\nu_S, \nu_T) - 2\epsilon$. All three lower bounds hold for every $f \in \mathcal{F}_{L,m,\mathrm{PE}}$, so $\mathcal{R}_{\mu_T}(f)$ is bounded below by their maximum. The same inequality from Lemma 1 gives the upper bound. □

E.2 Proof of Corollary 1

For completeness, we restate Corollary 1. Assume the source risk satisfies $\mathcal{R}_{\mu_S}(f_{\mathrm{RoPE}}) = \epsilon_{\mathrm{train}} \approx 0$, and suppose the target shift satisfies $W_1(\nu_S, \nu_T) = \Omega(T_{\max})$. Then the RoPE target risk degrades at most proportionally to the normalized shift: $\mathcal{R}_{\mu_T}(f_{\mathrm{RoPE}}) \leq \epsilon_{\mathrm{train}} + C_{\mathrm{smooth}}\, W_1(\nu_S, \nu_T)/T_{\max} + 2\epsilon \leq \epsilon_{\mathrm{train}} + C_{\mathrm{RoPE}} + 2\epsilon$, where $K_{f_{\mathrm{RoPE}}} \leq C_{\mathrm{smooth}}/T_{\max}$ (Theorem 1) and $C_{\mathrm{RoPE}} := C_{\mathrm{smooth}} \sup_{T_{\max}} \mathrm{diam}(\mathcal{X})/T_{\max} < \infty$ does not depend on $T_{\max}$.

In contrast, suppose there exists a measurable subset $\mathcal{X}_{\mathrm{shift}} \subset \mathcal{X}$ such that $\nu_T(\mathcal{X}_{\mathrm{shift}}) = p_{\mathrm{shift}} > 0$ and $\mathcal{L}_{f_{\mathrm{APE}}}(\boldsymbol{\chi}) \geq \gamma$ for all $\boldsymbol{\chi} \in \mathcal{X}_{\mathrm{shift}}$, where $\gamma = \Omega(1)$. Then $\mathcal{R}_{\mu_T}(f_{\mathrm{APE}}) \geq \gamma p_{\mathrm{shift}} - \epsilon$. In particular, if $\epsilon \ll \gamma p_{\mathrm{shift}}$, the APE target risk admits a non-vanishing lower bound.

Proof. For RoPE, Lemma 1 gives

$$\mathcal{R}_{\mu_T}(f_{\mathrm{RoPE}}) \leq \mathcal{R}_{\mu_S}(f_{\mathrm{RoPE}}) + K_{f_{\mathrm{RoPE}}} W_1(\nu_S, \nu_T) + 2\epsilon. \tag{97}$$

By Theorem 1, $K_{f_{\mathrm{RoPE}}} \leq C_{\mathrm{smooth}}/T_{\max}$, so $K_{f_{\mathrm{RoPE}}} W_1(\nu_S, \nu_T) \leq C_{\mathrm{smooth}}\, W_1(\nu_S, \nu_T)/T_{\max}$. Since $W_1(\nu_S, \nu_T)$ is bounded above by $\mathrm{diam}(\mathcal{X}) = \Theta(T_{\max})$, this term is at most C_{RoPE}. Substituting $\mathcal{R}_{\mu_S}(f_{\mathrm{RoPE}}) = \epsilon_{\mathrm{train}}$ gives both stated bounds.

For APE, Assumption 1 yields

$$\mathcal{R}_{\mu_T}(f_{\mathrm{APE}}) \geq \int_{\mathcal{X}} \mathcal{L}_{f_{\mathrm{APE}}}(\boldsymbol{\chi})\, d\nu_T(\boldsymbol{\chi}) - \epsilon. \tag{98}$$

Restricting the domain of integration to $\mathcal{X}_{\text{shift}}$ and using the assumed lower bound on $\mathcal{L}_{f_{\text{APE}}}$ gives

$$\mathcal{R}_{\mu_T}(f_{\text{APE}}) \geq \int_{\mathcal{X}_{\text{shift}}} \mathcal{L}_{f_{\text{APE}}}(\boldsymbol{\chi})\, d\nu_T(\boldsymbol{\chi}) - \epsilon \tag{99}$$

$$\geq \int_{\mathcal{X}_{\text{shift}}} \gamma\, d\nu_T(\boldsymbol{\chi}) - \epsilon \tag{100}$$

$$= \gamma\, \nu_T(\mathcal{X}_{\text{shift}}) - \epsilon \tag{101}$$

$$= \gamma p_{\text{shift}} - \epsilon. \tag{102}$$

□

E.3 Proof of Corollary 2

For completeness, we restate Corollary 2. Suppose the target measure assigns positive mass to trajectories whose structural depth exceeds the available implicit computation threshold: $p_{\text{deep}} := \nu_T(\{\boldsymbol{\chi} \in \mathcal{X} : \chi_1 > \alpha_{\text{circ}} L\}) > 0$. Then the depth bottleneck contributes the non-vanishing lower bound $\mathcal{R}_{\mu_T}(f) \geq C_{\text{depth}}\, p_{\text{deep}} - \epsilon$. In contrast, the width-based term $\frac{C_{\text{width}}}{\sqrt{m}} - \epsilon$ vanishes as $m \to \infty$. Moreover, if $L \geq \frac{1}{\alpha_{\text{circ}}} \sup_{\boldsymbol{\chi} \in \text{supp}(\nu_T)} \chi_1$, then $\nu_T(\{\boldsymbol{\chi} \in \mathcal{X} : \chi_1 > \alpha_{\text{circ}} L\}) = 0$, so the depth bottleneck term is eliminated on the support of ν_T.

Proof. By Lemma 5,

$$\mathcal{R}_{\mu_T}(f) \geq C_{\text{depth}} \int_{\mathcal{X}} P_{\text{implicit}}(\boldsymbol{\chi}; L, \alpha_{\text{circ}})\, d\nu_T(\boldsymbol{\chi}) - \epsilon. \tag{103}$$

Since $P_{\text{implicit}}(\boldsymbol{\chi}; L, \alpha_{\text{circ}}) = \mathbb{I}_{(\alpha_{\text{circ}} L, \infty)}(\chi_1)$, we have

$$\int_{\mathcal{X}} P_{\text{implicit}}(\boldsymbol{\chi}; L, \alpha_{\text{circ}})\, d\nu_T(\boldsymbol{\chi}) = \nu_T(\{\boldsymbol{\chi} \in \mathcal{X} : \chi_1 > \alpha_{\text{circ}} L\}) = p_{\text{deep}}, \tag{104}$$

which gives $\mathcal{R}_{\mu_T}(f) \geq C_{\text{depth}} p_{\text{deep}} - \epsilon$.

On the other hand,

$$\lim_{m \to \infty} \left(\frac{C_{\text{width}}}{\sqrt{m}} - \epsilon \right) = -\epsilon \leq 0, \tag{105}$$

so the width-only contribution cannot sustain a non-vanishing lower bound.

Finally, if $L \geq \frac{1}{\alpha_{\text{circ}}} \sup_{\boldsymbol{\chi} \in \text{supp}(\nu_T)} \chi_1$, then no point in $\text{supp}(\nu_T)$ satisfies $\chi_1 > \alpha_{\text{circ}} L$. Hence $\nu_T(\{\boldsymbol{\chi} \in \mathcal{X} : \chi_1 > \alpha_{\text{circ}} L\}) = 0$, which eliminates the depth bottleneck term on the target support. □

F Algebraic Equivalence Classes of Positional Encodings

Recent empirical studies frequently benchmark a wide array of positional encoding variants, including ALiBi [Press et al., 2022], T5 Relative Bias, and No Positional Encoding (NoPE) [Kazemnejad et al., 2023]. Our theoretical framework analyzes Absolute Positional Encoding (APE) and Rotary Position Embedding (RoPE) as representatives of two algebraic equivalence classes.

In the context of the structural Lipschitz gap (Theorem 1), the generalizability of an attention mechanism under sequence coordinate translations ($\mathcal{T}_k$) is determined by whether its attention kernel $A_{i,j}$ satisfies shift invariance. Positional encodings partition the architectural hypothesis space into two distinct algebraic equivalence classes:

1. Position-Dependent Attention Kernels: This class contains mechanisms where $A_{i,j}$ depends explicitly on the absolute indices i and j. Standard learned APE and sinusoidal encodings belong to this class. As proven in Lemma 2 for learned APE, members of this equivalence class fail to preserve shift invariance, causing the computational graph to differ under translations. Under Assumption 2, this yields a structural Lipschitz constant $K_f = \Omega(1)$.

2. Shift-Invariant Attention Mechanisms: This class contains mechanisms where the attention kernel depends only on the relative distance $i - j$. RoPE, ALiBi, and T5 Relative Bias all satisfy

this constraint. Consequently, for any mechanism in this equivalence class, the translation operator yields $\Delta A = 0$. Hence, any relative encoding in this class inherits the baseline structural smoothness $\mathcal{O}(1/T_{\max})$ established in Theorem 1.

Finally, No Positional Encoding (NoPE) injects no positional vectors, so its kernel satisfies $\Delta A^{\text{NoPE}} \equiv 0$ trivially, and the model recovers structural routing information implicitly through causal masking. Kazemnejad et al. [2023] show that such implicit recovery is possible (the first layer can reconstruct absolute positions) and that NoPE length-generalizes on par with or better than explicit encodings in their experiments. Its behavior under translations therefore depends on the positions recovered inside the network.

G Estimation of Structural Wasserstein Distances

The theoretical bounds in Theorem 2 are stated in terms of the exact Wasserstein-1 distance $W_1(\nu_S, \nu_T)$ between structural pushforward measures. In the empirical analysis, we estimate this quantity from finite samples by solving the exact (unregularized) optimal transport problem between empirical measures. Accordingly, we denote these finite-sample estimates by $\widehat{W}_1$, distinguishing them from the population-level W_1.

G.1 Computational Pipeline

Let $\hat{\nu}_S$ and $\hat{\nu}_T$ denote the empirical structural measures obtained by sampling trajectories from the source and target distributions and applying the structural projection Φ. The estimation pipeline consists of the following steps.

Joint standardization. The structural coordinates have different numerical scales; for example, the search-volume coordinate χ_4 can be orders of magnitude larger than transition-rate coordinates such as χ_5, χ_6, χ_7. To prevent the Euclidean cost from being dominated by large-scale coordinates, we apply Z-score normalization to each coordinate. The normalization statistics are computed once, on the pooled structural vectors of all three measures, so all pairs share one scale and their distances are directly comparable.

Cost matrix construction. For standardized structural vectors $\boldsymbol{u}_i, \boldsymbol{v}_j \in \mathbb{R}^{12}$, we define the empirical cost matrix by

$$M_{ij} = \|\boldsymbol{u}_i - \boldsymbol{v}_j\|_2. \tag{106}$$

For the entropically regularized cross-check below, we additionally rescale the cost matrix as

$$\widetilde{M}_{ij} = \frac{M_{ij}}{\max_{i,j} M_{ij} + 10^{-9}}. \tag{107}$$

This places the costs of the cross-check in $[0, 1]$.

Exact transport. With uniform marginal weights $\mathbf{p} = \mathbf{q} = N^{-1}\mathbf{1}$, we compute $\widehat{W}_1(\hat{\nu}_S, \hat{\nu}_T) = \min_{\mathbf{P} \in \Pi(\mathbf{p},\mathbf{q})} \langle \mathbf{P}, \mathbf{M} \rangle_F$ with the network-simplex solver of the Python Optimal Transport library [Flamary et al., 2021]. As a cross-check, the debiased Sinkhorn divergence (Sinkhorn–Knopp with $\lambda = 0.1$ on the rescaled cost $\widetilde{\mathbf{M}}$, Cuturi, 2013) preserves the separation reported below (within-measure $<$ BFS–MIXED $<$ BFS–DFS); the debiasing removes the entropic bias that dominates the uncorrected regularized cost at this sample size.

Subsampling. Each training measure contains 500,000 trajectories, making all-pairs cost-matrix construction expensive. For each distribution, we first form a candidate pool of 20,000 trajectories and then sample $N = 2{,}000$ trajectories without replacement for each pairwise distance computation. We repeat this procedure over 10 independent subsamples with fixed seeds and report the mean $\pm$ standard deviation.

G.2 Estimates and Interpretation

Across the 10 subsamples, the within-measure comparison (two disjoint samples of the same measure) remains small, while the MIXED and DFS target distributions produce much larger shifts:

$$\widehat{W}_1(\hat{\nu}_{BFS}, \hat{\nu}_{BFS}) = 0.19 \pm 0.06, \qquad \widehat{W}_1(\hat{\nu}_{BFS}, \hat{\nu}_{MIXED}) = 1.38 \pm 0.05,$$
$$\widehat{W}_1(\hat{\nu}_{BFS}, \hat{\nu}_{DFS}) = 2.97 \pm 0.05,$$

and $\widehat{W}_1(\hat{\nu}_{DFS}, \hat{\nu}_{DFS}) = 0.28 \pm 0.05$, $\widehat{W}_1(\hat{\nu}_{DFS}, \hat{\nu}_{MIXED}) = 1.76 \pm 0.06$. The separation between the three regimes, with BFS–MIXED and BFS–DFS distances 7 and 16 times the within-BFS baseline, shows that the structural feature space $\mathcal{X}$ captures the intended differences in search behavior between BFS, MIXED, and DFS trajectory distributions.

G.3 Empirical Calibration of the Theoretical Ceiling

In Figure 1A, we visualize the empirical accuracy of the evaluated architectures against the TC^0 depth capacity bottleneck. We now describe how the "Theoretical Ceiling" is motivated by Lemma 5 and calibrated.

By Theorem 2, the expected target risk on the DFS evaluation measure (μ_{DFS}) is lower-bounded by the depth penalty:

$$\mathcal{R}_{\mu_{DFS}}(f) \geq C_{\text{depth}} \int_{\mathcal{X}} P_{\text{implicit}}(\boldsymbol{\chi}; L, \alpha_{\text{circ}})\, d\nu_{DFS}(\boldsymbol{\chi}) - \epsilon. \tag{108}$$

Because empirical accuracy is defined as $1 - \mathcal{R}_{\mu_T}(f)$, this expected risk lower bound gives an empirical accuracy upper bound (the theoretical ceiling). Since χ_1 takes finitely many values (Appendix A.1), the truncated mass $\int_{\mathcal{X}} P_{\text{implicit}}\, d\nu_{DFS}$ is a non-increasing step function of the physical layer depth L, and we summarize its decay over the evaluated depths by a smooth exponential profile. On this task $\chi_1 \leq 4$ for every reference trajectory (Appendix K.5), so the curve below is a calibrated envelope, drawn for the DFS evaluation set, the hardest of the three.

We formulate the theoretical ceiling as:

$$\text{Accuracy}_{\max}(L) = \mathcal{A}_{\max} - \beta \cdot \exp(-\alpha \cdot L) \tag{109}$$

where $\beta, \alpha > 0$ are calibration constants and $\mathcal{A}_{\max}$ is the asymptotic empirical accuracy. The risk bound above constrains only the depth-dependent term and gives $\mathcal{A}_{\max} \leq 1 + \epsilon$, so we calibrate $\mathcal{A}_{\max}$ empirically, and it absorbs the failure sources outside the depth term. One of them is the generation budget: within the 512 new tokens allowed at evaluation (Appendix K), the reference trajectories themselves reach the target for 0.4645, 0.4860 and 0.4915 of the BFS, DFS and MIXED evaluation problems; most trained models, which are scored on their own searches, exceed these rates (up to 0.5360 on the DFS set), so the budget is a loose constraint on $\mathcal{A}_{\max}$.

We calibrated the constants ($\mathcal{A}_{\max} \approx 0.538, \beta \approx 0.035, \alpha \approx 0.15$) such that the curve acts as the strict supremum envelope for all 54 evaluated models. As shown in Figure 1A, every empirical data point lies strictly below this ceiling. The width conclusion rests on measured accuracies at fixed depth, independent of the fitted constants. Taking the best of the six models of each architecture on μ_{DFS} (three training measures, APE and RoPE), the gain from width falls below one point at the largest widths: at $L = 12$, widening from $d = 192$ to $d = 768$ adds 1.55 points ($0.5160 \to 0.5315$), but widening further to $d = 1536$, which quadruples the core parameters ($84.9\,\text{M} \to 339.7\,\text{M}$), changes accuracy by -0.2 points (0.5295); at $L = 24$, doubling d from 512 to 1024 adds 0.6 points ($0.5280 \to 0.5340$); and at $L = 6$, quadrupling d from 256 to 1024 adds 0.95 points ($0.5130 \to 0.5225$). Model by model, the two doublings that start from $d \geq 512$ ($L = 12$ and $L = 24$) lower APE accuracy on μ_{DFS} for every training measure (by 0.25–9.7 points) and change RoPE accuracy by -0.3 to $+1.3$ points. This confirms that, over the evaluated range, width scaling does not raise the ceiling.

H Supplementary Empirical Analyses

H.1 Mitigation by Translation Equivariance

In Section 5 of the main text, we stated that the structural smoothness provided by Rotary Position Embedding (RoPE) acts as a continuous structural regularizer, removing the extra depth that APE requires at Large scale on recursive backtracking (μ_{DFS}).

To isolate this effect, Figure 2 visualizes the performance trajectories of Large- and Medium-scale models trained and evaluated exclusively on the μ_{DFS} distribution.

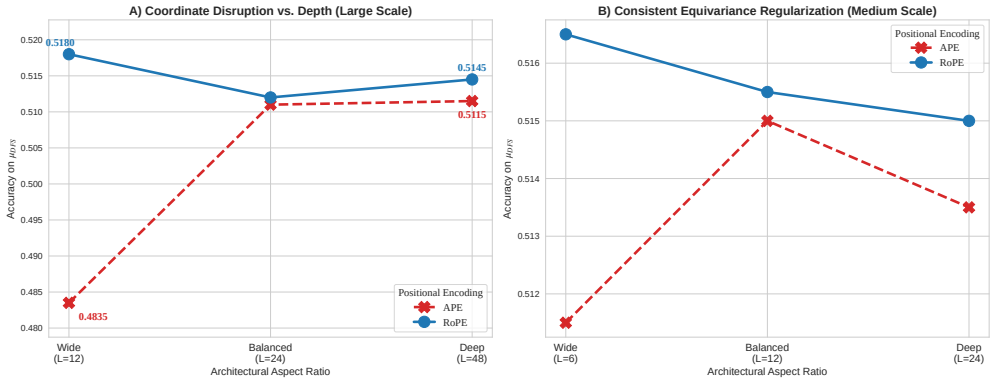


Figure 2: Empirical verification of structural regularization via translation equivariance on μ_{DFS}. **(A) Large Scale Models:** Under Absolute Positional Encoding (APE, red dashed line), increasing physical layer depth ($L = 12 \to L \geq 24$) raises accuracy from 0.4835 to 0.5110–0.5115. Because α_{circ} is the same for both encodings (Lemma 6) and Wide RoPE (blue solid line, $L = 12$) achieves 0.5180, 12 layers suffice on this task. We therefore attribute the failure of Wide APE ($L = 12$, 0.4835) to its coordinate disruption (Lemma 2, Assumption 2): at this scale APE needs additional physical depth ($L \geq 24$) to memorize absolute-position-dependent routing, whereas RoPE, which preserves translation equivariance, succeeds with 12 layers. **(B) Medium Scale Models:** The same comparison at $L = 6$, 12 and 24 (Wide, Balanced, Deep): APE reaches 0.5115, 0.5150 and 0.5135, and RoPE 0.5165, 0.5155 and 0.5150, so RoPE lies above APE for every shape, by at most 0.5 points.

At Large scale (Panel A), APE accuracy (red dashed line) rises monotonically with depth, from 0.4835 (Wide, $L = 12$) to 0.5110 ($L = 24$) and 0.5115 ($L = 48$); the same monotone rise holds at this scale for every combination of training measure and evaluation set. Twelve layers nonetheless suffice on this task: Wide RoPE reaches 0.5180 at the same $L = 12$, and the Medium APE models of Panel B reach 0.5115 at $L = 6$ and 0.5150 at $L = 12$. The added depth thus compensates for a failure specific to the widest APE model, which the next paragraph attributes to coordinate disruption.

In contrast, RoPE flattens this trajectory. As proven in Lemma 6, RoPE and APE possess identical theoretical expressivity limits (α_{circ}). However, without relative equivariance, APE architectures are limited by coordinate disruption (Assumption 2; Corollary 1) before reaching their capacity, so that at Large scale APE needs greater depths ($L \geq 24$) to provide implicit routing redundancy. By preserving structural shift invariance, RoPE smooths the risk landscape over $\mathcal{X}$ and avoids these coordinate disruptions. The Large Wide DFS RoPE configuration achieves 0.5180, on par with 0.5145 for its Deep counterpart. This visual evidence provides empirical support for our theoretical claim: translation equivariance acts as a structural regularizer that lets the architecture reach its expressivity bounds without premature collapse.

H.2 Baseline Capacity and Structural Variance

Figure 3 gives an aggregate view of in-distribution performance ($\nu_T = \nu_S$) across all three parameter scales and training measures.

The main observation from this aggregate view is the presence of *structural variance*, denoted by the standard deviation error bars. Because each bar aggregates the performance of the Deep, Wide, and Balanced variants within a matched core-parameter budget, the magnitude of the error bar reflects the model's sensitivity to the computational graph's aspect ratio.

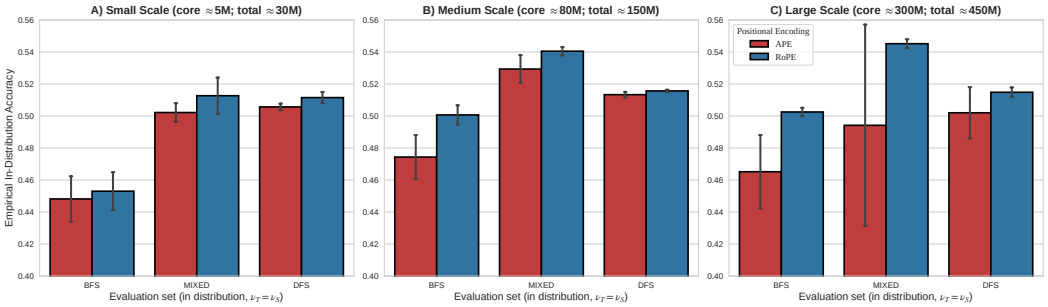


Figure 3: Baseline performance and structural variance across the experimental grid at Small (A), Medium (B) and Large (C) scale. For in-distribution evaluation ($\nu_T = \nu_S$), the bars represent the mean empirical accuracy across different architectural aspect ratios (Deep, Wide, Balanced), while the error bars denote ± 1 standard deviation. The large variance under APE at the Large scale (largest on μ_{MIXED}) shows that aspect ratio dominates performance, whereas RoPE reduces this variance at the Medium and Large scales, supporting its role as a structural regularizer.

As evident in Figure 3, the structural variance under Absolute Positional Encoding (APE, red bars) is high, particularly at the Large scale, where it is driven by the Wide architecture (most visibly its collapse under μ_{MIXED}, Figure 1C). This high variance shows that under absolute sequence coordinates, accuracy at a matched parameter budget is highly sensitive to the aspect ratio (at Large scale, APE accuracy rises with depth for every training measure, by up to 11.15 points in distribution).

By contrast, Rotary Position Embedding (RoPE, blue bars) has smaller error bars at the Medium and Large scales. Its translation equivariance reduces the architecture's sensitivity to the discrete layout of the computational graph. This aggregated view supports our theoretical finding: preserving local structural equivariance acts as a continuous structural regularizer, raising the lowest accuracy across depth-to-width ratios in eight of the nine scale–measure cells (all but Small-scale BFS, where the two differ by 0.05 points).

I Further Intuition on the Bounds

What bounds the accuracy of RoPE models. RoPE removes the APE-specific coordinate-disruption failure (Corollary 1) and, under a measure shift, controls the Lipschitz penalty $K_{f,\mathrm{PE}}\, W_1(\nu_S, \nu_T)$ of Theorem 2; the two capacity terms, the depth term $C_{\mathrm{depth}} \int P_{\mathrm{implicit}}\, d\nu_T$ and the width term $C_{\mathrm{width}}/\sqrt{m}$, are the same for both encodings. A shift-invariant architecture therefore avoids the collapse caused by coordinate shifts, while its absolute accuracy remains bounded by the depth and width terms and by the residual ϵ. Accordingly, the recovery from 0.4215 (Large Wide MIXED APE) to 0.5425 (its RoPE counterpart) brings the model to the level of the other Large architectures. That level is saturated in depth and width over the evaluated range: on each evaluation set the best accuracies at $L = 12$, 24 and 48 lie within one point of each other, and the width comparison of Appendix G.3 shows gains below one point at the largest widths.

Controlling ϵ and the dimension of $\mathcal{X}$. Enlarging $\mathcal{X}$ with finer descriptors (e.g., token identities, n-gram statistics, operand values) would drive the sufficiency residual ϵ of Assumption 1 toward zero, at the price of a sample complexity for estimating W_1 that grows rapidly with the dimension [Weed and Bach, 2019]. The twelve structural coordinates balance a residual ϵ against stable estimation of the structural shift (Appendix G).

How K_f enters the experiments. K_f enters through its theoretical bounds: $K_{f,\mathrm{APE}} = \Omega(1)$ and $K_{f,\mathrm{RoPE}} = \mathcal{O}(1/T_{\max})$ follow from Assumption 2 (Lemma 2 and Theorem 1), and Section 5 examines their empirical consequences across architectures and training measures.

J Empirical Check of Structural Sufficiency (Assumption 1)

We check Assumption 1 for two trained models by predicting their per-problem loss from $\boldsymbol{\chi}$, within and across the three search measures. All metrics are computed out of fold on freshly generated trajectories (50,000 for one model and 3,000 of them for the other) and compared with explicit baselines.

Setup. The label is the zero-one loss $\ell(f, z)$ of Appendix A.4: whether model f, decoding greedily with the settings of Appendix K, fails on the problem $x_{\text{init}}^{(z)}$ of the reference trajectory z. The model is prompted with this problem only, so ℓ is a deterministic function of it (repeated prompts always received the same label). Data: 50,000 newly generated reference trajectories (generator seed 2026; 16,667/16,667/16,666 from $\mu_{BFS}, \mu_{DFS}, \mu_{MIXED}$). Of their problems, 37.7% (39.3% of the subset used for f_2) also occur in at least one training set, almost all of them with 3 operands; Table 2 therefore also reports the problems absent from all training sets. Models: Small Deep MIXED RoPE (f_1, all 50,000 trajectories) and Large Deep BFS APE (f_2, the first 1,000 per measure), two of the 54 configurations of Section 5, which differ in scale, training measure and positional encoding. Every input is computed from the reference trajectory z, never from the model's output: $\boldsymbol{\chi} = \Phi(z)$; $\boldsymbol{\chi}$ without the reference solution indicator χ_{10}; the token length of z (LEN); the problem alone (PROB: target, operand count and sorted operands, i.e., the prompt up to operand order); and the operand count alone. $\hat{\mathcal{L}}$ is scikit-learn's `HistGradientBoostingClassifier` [Pedregosa et al., 2011] with default settings, evaluated out of fold under stratified 5-fold cross-validation. McFadden's pseudo-R^2 is $1 - \log L_{\hat{\mathcal{L}}} / \log L_0$, where L_0 is the likelihood of a null model that predicts the training-fold base rate [McFadden, 1979]. The code and the per-problem labels are available on GitHub (Appendix L). The classifier fits $P(\ell{=}0)$, as in Table 2; with $P(\ell{=}1)$ the fits differ only through the default early stopping, which is active for the training sets of f_1 (more than 10,000 samples) and draws a validation split that depends on the coding. Switching the coding changes every reported value by at most 0.003, except the risk estimates of f_1 classifiers fitted on BFS or DFS trajectories alone (up to 0.04), which we report below under $P(\ell{=}1)$, with the value under $P(\ell{=}0)$ in parentheses.

Table 2: Per-trajectory prediction of ℓ by $\hat{\mathcal{L}}$: out-of-fold ROC-AUC and McFadden pseudo-R^2 (mean over five folds; $\pm$ population s.d. over folds for $\boldsymbol{\chi}$). "Not in training": problems absent from all training sets (n = 31,173 for f_1 and 1,822 for f_2; mostly with 4 operands). "Within μ": cross-validated within the trajectories of one search measure (16,667/16,667/16,666 for f_1, 1,000 each for f_2).

	f_1 (n = 50,000)		f_2 (n = 3,000)	
Input	ROC-AUC	McFadden	ROC-AUC	McFadden
$\boldsymbol{\chi}$	0.927 ± 0.002	0.51 ± 0.01	0.929 ± 0.007	0.48 ± 0.03
$\boldsymbol{\chi}$ without χ_{10}	0.926	0.50	0.927	0.47
LEN	0.903	0.44	0.917	0.46
PROB	0.890	0.40	0.906	0.41
Operand count only	0.797	0.27	0.853	0.40
$\boldsymbol{\chi}$, not in training	0.905	0.44	0.910	0.36
$\boldsymbol{\chi}$, within μ_{BFS}	0.905	0.44	0.940	0.49
$\boldsymbol{\chi}$, within μ_{DFS}	0.952	0.60	0.912	0.39
$\boldsymbol{\chi}$, within μ_{MIXED}	0.918	0.48	0.919	0.41

Per trajectory. Out of fold, $\boldsymbol{\chi}$ predicts ℓ with ROC-AUC 0.927 and McFadden pseudo-R^2 0.51 for f_1 (0.929 and 0.48 for f_2); removing χ_{10} changes little. McFadden's index runs considerably lower than the R^2 of linear regression, and values of 0.2–0.4 already represent an excellent fit [McFadden, 1979]. Every input in Table 2 lies in or above that range, and among the inputs evaluated on the full sample, $\boldsymbol{\chi}$ scores highest for both models. For f_1, $\boldsymbol{\chi}$ exceeds LEN and PROB on both metrics in every fold. For f_2, it exceeds them in ROC-AUC in every fold and in mean McFadden value (0.48 against 0.41 for PROB and 0.46 for LEN, which it exceeds in three of five folds). Problem size carries a large part of the signal (accuracy is about 0.8 with 3 operands and 0.1–0.2 with 4), and $\boldsymbol{\chi}$ goes beyond it: ROC-AUC 0.927 against 0.797 and McFadden 0.51 against 0.27 for the operand count alone (f_1), and 0.929 against 0.853 and 0.48 against 0.40 (f_2). On problems absent from all

training sets, χ remains strongly predictive (ROC-AUC 0.905 and 0.910, McFadden 0.44 and 0.36 for f_1 and f_2). Within a single search measure, χ reaches ROC-AUC 0.905–0.952 (f_1) and 0.912–0.940 (f_2). PROB is nearly the whole prompt, which determines ℓ, yet scores below χ: the classifier extracts more predictive signal from the structural coordinates than from the problem itself.

Across search measures. The three search measures share one problem distribution and differ only in the reference search trace, which the model never sees (Appendix K.5). On freshly generated problems, a fixed model therefore has the same risk under all three measures up to sampling noise ($0.502/0.494/0.500$ for f_1, $0.511/0.511/0.490$ for f_2). Assumption 1 thus holds for every pair of these measures with ϵ equal to the largest gap between a measure's risk and the pooled risk (0.004 for f_1, 0.014 for f_2), attained by the constant $\mathcal{L}_f$ equal to the pooled risk; a constant fitted on one measure misestimates the risk of another by at most 0.007 (f_1) and 0.021 (f_2). A transfer analysis fits $\hat{\mathcal{L}}$ on all trajectories of one search measure and applies it to another: it ranks the target trajectories with ROC-AUC 0.87–0.95, and its mean prediction matches the target risk within 0.043 (0.061) for f_1 and 0.038 for f_2. We take this transfer error as the empirical counterpart of ϵ; it measures how stable the χ-conditional risk is across search procedures.

Sub-populations. We group the trajectories into 54 (f_1) and 18 (f_2) cells of search measure $\times$ operand count $\times$ target band; unlike the three search measures, cells that differ in operand count or target band pose different problems. In each cell we compare the observed risk with the mean cross-fitted prediction of $\hat{\mathcal{L}}$ on χ, fitted on trajectories from all cells: the absolute deviation averages 0.031 (maximum 0.114) for f_1 and 0.038 (maximum 0.081) for f_2, against 0.297 and 0.353 for a constant (LEN: 0.030 and 0.048; PROB: 0.010 and 0.016). If $\hat{\mathcal{L}}$ were perfectly calibrated (labels simulated from its own predictions), the largest deviation would stay below 0.037 (f_1) and 0.070 (f_2) in 95% of samples; both observed maxima exceed these levels. Fitted on a single search measure and applied to the cells of the other two, $\hat{\mathcal{L}}$ deviates from the cell risk by at most 0.196 (0.237) for f_1 and 0.116 for f_2. Across cells, the cross-fitted predictions of $\hat{\mathcal{L}}$ on χ reach a coefficient of determination of $0.983/0.985$ (f_1/f_2) for the observed risks, against about 0 for a constant; because the cells are defined by problem difficulty, simpler summaries reach comparable or higher values ($0.986/0.977$ for LEN, $0.998/0.996$ for PROB, $0.969/0.964$ for the operand count alone).

Conclusion. χ is a strong predictor of per-problem success: by ROC-AUC it ranks above LEN, PROB and the operand count for both models, and it stays predictive on problems absent from all training sets. Across the three search measures, which share one problem distribution, a fixed model has the same risk up to sampling noise, so Assumption 1 holds with $\epsilon = 0.004$ (f_1) and 0.014 (f_2).

K Experimental Details and Reproducibility

To support the reproducibility of our empirical results and structural metrics, we detail the exact computational environment, architectural implementations, and optimization regimens corresponding to the 54 Transformer configurations evaluated in Section 5.

K.1 Compute Infrastructure and Environment

All experiments, including dataset generation, model training, and trajectory evaluations, were executed on a high-performance Linux cluster. The hardware infrastructure comprises nodes equipped with 4 or 8 NVIDIA A100 (40 GB) Tensor Core GPUs. Job scheduling was managed via Slurm.

Each individual model training run was allocated a dedicated environment consisting of 1 NVIDIA A100 GPU, 16 CPU cores, and 128 GB of system RAM, with a maximum wall-clock time limit of 7 days; training took 2.7–4.6 h (Small), 8.2–21.9 h (Medium) and 18.3–58.7 h (Large), about 907 GPU-hours in total. The evaluation pipeline was allocated 1 NVIDIA A100 GPU, 24 CPU cores, and 128 GB of system RAM (about 324 GPU-hours in total). Labeling the problems of Appendix J by greedy decoding took about 17 further GPU-hours on the same hardware (single-GPU jobs with 8 CPU cores and 48 GB of RAM); the remaining computations (generating the problems of Appendix J, the $\widehat{W}_1$ estimates of Appendix G and the analyses of Appendix J) used CPUs only. For Appendix M, Experiment 3 took about 23.5 GPU-hours on NVIDIA A100 (80 GB) GPUs, with grading on CPUs, Experiment 4 about 0.45 GPU-hours on NVIDIA H100 (80 GB) GPUs, Experiment 5 about 106 GPU-hours on NVIDIA H100 and L40S GPUs, Experiment 6 about 4.5 GPU-hours on NVIDIA

A100 (80 GB) and H100 GPUs, and Experiments 7 and 8 about 119 and 13 GPU-hours on NVIDIA A100 and L40S GPUs. Experiments 1 and 2 used CPUs only.

The software environment was built upon Python 3.10, PyTorch 2.5.1 (CUDA 12.4), Hugging Face `transformers` 4.57.1, `datasets` 4.4.1 and `accelerate` 1.12.0; attention used the PyTorch SDPA kernel for GPT-NeoX and the eager implementation for GPT-Neo. During evaluation, the PyTorch cuDNN backend was set to deterministic mode (`torch.backends.cudnn.deterministic = True`).

Retraining check. Retraining the 18 Small models of Section 5 from scratch with the released training code and configuration, on training data regenerated by the released generator, reproduces their accuracies on the three evaluation sets within 0.02 (mean absolute difference 0.006 over the 54 model–set pairs); the retraining ran on NVIDIA L40S and A100 GPUs and took about 57 GPU-hours.

K.2 Architectural Implementations

To isolate the positional encoding variable ($\mathrm{PE} \in \{\mathrm{APE}, \mathrm{RoPE}\}$), we mapped the architectures to their respective native implementations in the Hugging Face library:

- **APE Configurations:** Evaluated using the `GPTNeoForCausalLM` architecture. The absolute positional embeddings were configured up to the maximum context length (1024). Global attention was enforced in all layers (i.e., `attention_types=[[["global"], n_layer]]`) to prevent confounding effects from sliding-window or local attention approximations.
- **RoPE Configurations:** Evaluated using the `GPTNeoXForCausalLM` architecture, which natively supports rotary mappings. We set the rotary percentage to 1.0 (applying RoPE to all attention head dimensions) and disabled parallel residual connections (`use_parallel_residual=False`) to keep the computational graph as close as possible to the APE baseline, isolating translation equivariance as the principal independent variable (the two implementations retain minor differences, e.g., attention-logit scaling and input/output embedding tying).

Both architectural families used GELU activations (the exact GELU in GPT-NeoX and its tanh approximation, `gelu_new`, in GPT-Neo) and vocabulary dimension $|\mathcal{V}| = 50{,}257$. During training of the Medium and Large models, memory use was reduced by enabling gradient checkpointing (`gradient_checkpointing=True`), which implicitly disables key-value caching during the forward pass. Computations were uniformly executed in `bfloat16` mixed precision to prevent numerical overflow while accelerating matrix multiplications.

K.3 Optimization Regimen

All 54 configurations were trained from scratch under a single shared hyperparameter configuration for controlled capacity comparisons:

- **Data & Context:** The models were trained on 500,000 trajectories per training measure. Sequences were truncated or padded to $T_{\max} = 1024$.
- **Loss Function:** Standard causal language modeling (next-token prediction) using cross-entropy loss. Padding tokens were ignored during loss computation by overriding their labels to -100; because the padding token is the end-of-text token that also opens and closes each trajectory, these delimiters were excluded from the loss as well.
- **Batch Size:** The effective batch size was 32 trajectories per optimization step across all runs: a per-device batch of 32 for the Small models, and a per-device batch of 4 with 8 gradient accumulation steps for the Medium and Large models.
- **Learning Rate & Optimizer:** We used the AdamW optimizer with a single peak learning rate of $\eta = 8 \times 10^{-5}$ across all scales (Small, Medium, Large). The learning rate followed a linear warmup schedule for the first $1{,}000$ steps and then decayed linearly to zero. Weight decay was uniformly set to 0.01.

- **Epochs & Seeds:** All models were trained for exactly 3 full epochs. Results for each configuration come from a single training run, with the global random seed for data shuffling and weight initialization fixed to 42 (`random.seed(42)`, `torch.manual_seed(42)`).

K.4 The Countdown Task and the Stream of Search Format

In Countdown, a set of input numbers must be combined with the operations $\{+, -, \times, \div\}$, each number used once, to reach a target. Consider the target 10 with numbers $[2, 3, 5]$. The depth-first search below first expands $2 \times 5 = 10$ (remaining $[3, 10]$), whose children $3+10 = 13$ and $10-3 = 7$ miss the target; it backtracks to the root, expands $2 \times 3 = 6$ (remaining $[5, 6]$), again meets only dead ends (11 and 1), backtracks once more, and finally expands $2 + 3 = 5$ (remaining $[5, 5]$) and $5 + 5 = 10$, which reaches the target. Stream of Search [Gandhi et al., 2024] serializes such a tree, including every dead end and backtracking step, into a single token sequence. The trajectory below is the one produced by the depth-first generator used for μ_{DFS} on this instance (sum heuristic, pruning threshold equal to the target); the model is prompted with the first line and must generate the rest.

```
Current State: 10:[2, 3, 5], Operations: []
Exploring Operation: 2*5=10, Resulting Numbers: [3, 10]
Generated Node #0,0: 10:[3, 10] Operation: 2*5=10
Moving to Node #0,0
Current State: 10:[3, 10], Operations: ['2*5=10']
Exploring Operation: 3+10=13, Resulting Numbers: [13]
13,10 unequal: No Solution
Moving to Node #0,0
Current State: 10:[3, 10], Operations: ['2*5=10']
Exploring Operation: 10-3=7, Resulting Numbers: [7]
7,10 unequal: No Solution
Moving to Node #0
Current State: 10:[2, 3, 5], Operations: []
Exploring Operation: 2*3=6, Resulting Numbers: [5, 6]
Generated Node #0,1: 10:[5, 6] Operation: 2*3=6
...                          (two dead ends below node #0,1, then backtrack to #0)
Exploring Operation: 2+3=5, Resulting Numbers: [5, 5]
Generated Node #0,2: 10:[5, 5] Operation: 2+3=5
Moving to Node #0,2
Current State: 10:[5, 5], Operations: ['2+3=5']
Exploring Operation: 5+5=10, Resulting Numbers: [10]
10,10 equal: Goal Reached
```

K.5 Evaluation Procedure

During the empirical evaluation phase, the models were evaluated on sets of 2,000 trajectories per structural measure ($\nu_{BFS}, \nu_{DFS}, \nu_{MIXED}$). For ν_{BFS} and ν_{DFS} these are the first 2,000 trajectories generated for the corresponding training measure (the generator is seeded identically), so in-distribution scores measure how well a model fits its training measure; ν_{MIXED} combines 1,000 trajectories from each. The autoregressive generation was prompted identically using the initial problem state: `<|endoftext|>Current State: [Target]:[Nums], Operations: []` followed by a newline.

To assess the structural fidelity of the learned models without stochastic sampling variance, we used greedy decoding (`do_sample=False`, `num_beams=1`) with a maximum new token generation length of 512. Evaluation used a batch size of 32 with left padding for RoPE models and a batch size of 1 (no padding) for APE models. The generated sequences were subsequently parsed by the deterministic scoring function of the original Stream of Search implementation [Gandhi et al., 2024], which checks the arithmetic of the reported operations and whether the target state is reached, yielding the expected zero-one risk $\mathcal{R}_{\mu_T}(f)$ reported in Section 5.

Comparisons across measures. The three measures differ only in the search procedure that serializes the reference trajectory: the generator draws the number of operands, the target and the operands in the same way for all three, so they share one problem distribution. Because the model is prompted with the problem alone and the scorer checks only the model's own output, a fixed model has the same risk under μ_{BFS}, μ_{DFS} and μ_{MIXED}. Their structural measures, however, are

far apart (Appendix G): the DFS reference trajectories backtrack nearly four times as often as the BFS ones (mean $\chi_6 = 0.18$ against 0.05 on the evaluation sets), with tree depth $\chi_1 \leq 3$ against $\chi_1 \leq 4$. On the freshly generated problems of Appendix J, Large Deep BFS APE has risk 0.511, 0.511 and 0.490 under the three measures, whereas its accuracy on the evaluation sets falls from 0.4815 (BFS) to 0.4555 (DFS). Differences between the evaluation sets therefore reflect their particular $2{,}000$-problem samples. Averaged over the 18 models of each training measure, accuracy on the BFS, MIXED and DFS sets is 0.4740, 0.4699 and 0.4513 (BFS-trained), 0.5132, 0.5214 and 0.5105 (DFS-trained), and 0.5175, 0.5207 and 0.5110 (MIXED-trained): the DFS set is the hardest for every training measure, and the drop from the BFS to the DFS set is larger for BFS-trained models (2.27 points, against 0.27 and 0.65 for DFS- and MIXED-trained models). In comparisons of architectures and training measures on a common evaluation set (Section 5), every model is scored on the same problems.

L Code Availability

Code for data generation, training, evaluation and analysis is available at `https://github.com/Yyuzrah/AMTA4R`.

M Empirical Checks of the Terms of the Main Bound

On Countdown, the projection Φ of Theorem 2 maps a trajectory to the twelve coordinates $\boldsymbol{\chi}$ of Appendix A.1. Appendix M.1 derives from the terms of the bound what the coordinates of a projection must satisfy, and we then report eight experiments that we designed. Experiment 1 examines whether more coordinates improve the projection on Countdown. Experiments 2 and 3 ask whether $\widehat{W}_1$ separates groups of agent, embodied and robot trajectories whose tasks differ in structure, and whether a change of agent program alone shifts the trajectories and the success of one model (Appendix M.2). Experiments 4–6 test Lemma 2, which underlies the factor $K_{f,\mathrm{PE}}$, on attention scores, on answers under shifted position indices and on two Countdown models (Appendix M.3). Experiments 7 and 8 concern the depth term and whether splitting a deeply nested task into separate model calls restores the success that one call loses (Appendix M.4). The code repository (`https://github.com/Yyuzrah/AMTA4R`) holds the code, all per-pair, per-model and per-run results, and a catalogue of candidate coordinates for Countdown and six other trajectory domains.

M.1 How the Coordinates of Φ Are Chosen

A projection Φ_D for a trajectory domain D maps a trajectory to d_Φ coordinates. The upper bound of Theorem 2 adds to the source risk the shift term $K_{f,\mathrm{PE}} W_1(\nu_S, \nu_T)$ and twice the residual ϵ of Assumption 1, where ν_S and ν_T are the distributions of $\Phi(z)$ on the source and target tasks and $K_{f,\mathrm{PE}}$ is the Lipschitz constant of the risk as a function of $\Phi(z)$. The lower bound is the largest of three terms: the source risk minus the shift term and 2ϵ, a width term $C_{\mathrm{width}}/\sqrt{m} - \epsilon$ for a model of width m, and a depth term $C_{\mathrm{depth}} \int P_{\mathrm{implicit}}\, d\nu_T - \epsilon$. Every term except the width term places a requirement on the coordinates, and so does the estimation of W_1 from samples. We state these requirements in terms of events, the steps of a trajectory, such as the actions and observations of an agent or the visited nodes of a Countdown search.

The residual ϵ holds what $\Phi(z)$ leaves out, such as the operand values of a Countdown problem (Appendix A.2). A coordinate therefore depends on the kinds of events and on their positions in the sequence of events, and leaves content such as operands or object names to ϵ. Since ν_S and ν_T describe tasks, every coordinate is computed from the reference trajectory, the trajectory of a reference solver for the same problem, and may include the reference's own outcome, such as whether the reference search reached the target (χ_{10}). The shift term needs Φ to be bounded, so each coordinate is finite and grows at most linearly with the largest trajectory length $T_{\max}$. Because $K_{f,\mathrm{PE}}$ is taken in the Euclidean distance, a unit step along a coordinate must mean the same everywhere. Categories therefore enter as counts, one per category, as the Countdown operators enter through χ_{11} (multiplications and divisions) and χ_{12} (additions and subtractions), and any weighting of the coordinates is the same for every compared measure.

The gap $K_{f,\mathrm{APE}} \gg K_{f,\mathrm{RoPE}}$ of Theorem 1 uses the operator $\mathcal{T}_k$ of Definition 2, which inserts one event that changes no state, such as the identity node `x=x`, written with k tokens. The proof bounds $K_{f,\mathrm{APE}}$ from below by the jump γ in risk divided by the distance δ by which the insertion moves $\Phi(z)$, so δ must be positive and bounded in k and $T_{\max}$. Every coordinate therefore changes under $\mathcal{T}_k$ by a bounded amount, and one coordinate counts the inserted event. The search volume χ_4, the number of node references, grows by one whatever k is, whereas the token count of the trajectory grows by k and would let γ/δ vanish for long insertions. The depth term needs one coordinate that counts the nesting that one forward pass must resolve (Assumption 5), on Countdown $\chi_1 - 1$, the nesting depth of the bracket word of Lemma 3. The set of coordinates is small, because the number of samples needed to estimate W_1 grows quickly with the dimension [Weed and Bach, 2019].

Lemmas 1 and 4 and Assumption 4 are formulated on a general space $\mathcal{X}$, so Theorem 2, stated on $\mathbb{R}^{d_\Phi}$, keeps its form for any projection Φ_D with these properties, provided its assumptions hold in the domain. The twelve coordinates $\boldsymbol{\chi}$ have them: they are counts, rates and depths of events of the reference trajectory, the coordinates other than χ_4 change under $\mathcal{T}_k$ by an amount of order $1/\chi_4$ or not at all (Appendix C.1), and Appendix J measures how well they predict success.

M.1.1 Experiment 1: More Coordinates on Countdown

Adding coordinates can only raise W_1, since a distance over more coordinates is at least as long, and can only lower the smallest attainable ϵ, since a risk function of the old coordinates remains available. The upper bound becomes tighter only if 2ϵ falls by more than $K_{f,\mathrm{PE}} W_1$ grows, and W_1 is harder to estimate in more dimensions [Weed and Bach, 2019]. Experiment 1 measures both effects on Countdown with four nested projections $\Phi_{12} = \boldsymbol{\chi} \subset \Phi_{16} \subset \Phi_{24} \subset \Phi_{34}$, where the subscript is the number of coordinates. The 22 further coordinates are computed on the reference trajectory and enter class by class, in the order of the catalogue in the code repository. Φ_{16} adds four coordinates of the shape of the search tree, such as the revisit ratio, Φ_{24} eight of how the search moves and what it must remember, such as the longest streak of failures, and Φ_{34} ten more of memory, nesting, actions and length, such as the number of lines. Four of the further coordinates, the token length among them, are counted in tokens; the two measurements concern ϵ and W_1, and the bounded change under $\mathcal{T}_k$ enters only the gap of Theorem 1.

The first measurement repeats for each projection the prediction of success of Appendix J, in which a gradient-boosting classifier predicts from the coordinates of z whether a trained model fails on the problem of z. It uses the two models of that appendix, f_1 (Small Deep MIXED RoPE, 12 layers) with outcomes on $50{,}000$ problems and f_2 (Large Deep BFS APE, 48 layers) with outcomes on $3{,}000$ problems. We report the out-of-fold ROC-AUC and McFadden pseudo-R^2 over ten repetitions of stratified five-fold cross-validation, with the same folds for all projections. The second measurement estimates $\widehat{W}_1$ between the training trajectories of μ_{BFS}, μ_{DFS} and μ_{MIXED}, together with the within-measure baselines, as in Appendix G.

From Φ_{12} to Φ_{34} the ROC-AUC changes by at most 0.003, staying between 0.927 and 0.929 for both models, and the McFadden pseudo-R^2 by at most 0.016. The within-measure baselines grow by a factor of 1.6–2.3 and the between-measure distances by 1.8–2.0, for example from 0.19 to 0.44 for the baseline of μ_{BFS} and from 2.97 to 5.65 between μ_{BFS} and μ_{DFS}. The ratio of the mean baseline to the mean between-measure distance stays at 0.13. On Countdown the further coordinates thus roughly double $\widehat{W}_1$ and its noise floor without improving the prediction of success, and the twelve coordinates are the better choice. The same two measurements can be run in any domain to decide whether a coordinate is worth adding.

M.2 Agent, Embodied and Robot Trajectories

Experiments 2 and 3 apply projections Φ_D and the estimate $\widehat{W}_1$ to recorded logs of language agents, code agents and robots. Experiment 2 compares groups of reference trajectories, and Experiment 3 runs one language model in two agent harnesses, the programs that build the prompt, offer the model its tools and run its commands.

M.2.1 Experiment 2: Structural Shift between Groups of Trajectories

Experiment 2 tests whether $\widehat{W}_1$ reports a clear shift between groups of trajectories whose tasks differ in structure, and a small one between groups whose tasks share it. We use reference trajectories from four public sources. The first are the expert trajectories released with ETO [Song et al., 2024a,b], a method for training language agents, in ALFWorld [Shridhar et al., 2021] (household tasks, 3,119 trajectories), ScienceWorld [Wang et al., 2022] (science experiments, 1,483) and WebShop [Yao et al., 2022] (shopping, 1,824, of which 905 by humans and 919 by GPT-4). The second are two sets of code-agent runs on GitHub issues of SWE-rebench [Badertdinov et al., 2025], released by Nebius: 12,467 resolved runs on 835 issues of a 70B Llama model (`swe-agent-llama-70b`) in the SWE-agent harness [Yang et al., 2024, Nebius, 2024], and 32,161 on 3,792 issues of Qwen3-Coder-480B-A35B-Instruct [Qwen Team, 2025] in the OpenHands harness [Wang et al., 2025, Trofimova et al., 2025]. The third are the 33,252 teleoperated robot-arm demonstrations of BridgeData V2 [Walke et al., 2023] with a non-empty instruction, with the per-step movement primitives released with ECoT [Zawalski et al., 2025, Embodied-CoT, 2024]. The fourth are 4,000 levels per instruction kind of the grid world BabyAI [Chevalier-Boisvert et al., 2019], generated with BALROG [Paglieri et al., 2024] and Minigrid [Chevalier-Boisvert et al., 2023] and solved by the BabyAI bot.

For each source, a parser turns each log into a sequence of events and marks which events open and close a level of nesting, such as taking and putting down an object in ALFWorld or the explicit subgoal stack of the BabyAI bot. Each projection has 12 to 16 coordinates, among them the maximum and mean depth, counts of steps by type and, depending on the domain, revisits and failures. All projections except the code-agent one include a few surface coordinates (S), totals such as the number of actions.

We form 52 pairs of groups in six families: the six ALFWorld task types (15 pairs), the three ETO environments (3), human against GPT-4 demonstrations in WebShop (1), SWE-agent against OpenHands on the 554 issues resolved by both and on all resolved issues (2), seven BridgeData skills (21) and five BabyAI instruction kinds (10). $\widehat{W}_1$ is computed as on Countdown (Appendix G), with the coordinates z-scored on the pooled family and samples of $N = 2{,}000$ trajectories, or half the smaller group if that is less (139–2,000). The baseline of a group is $\widehat{W}_1$ between two disjoint samples of that group, and the ratio R divides the mean between-group distance over ten seeds by the larger mean baseline. A pair is separated when $R \geq 3$ and the between-group $\widehat{W}_1$ has a coefficient of variation over seeds of at most 0.10.

Figure 4 shows R for all pairs. Of the 52 pairs, 33 are separated, and all coefficients of variation are at most 0.095. Separated are the three ETO environments ($R = 6.26$–10.60), nine of the ten BabyAI pairs (10.94–28.65), the human and GPT-4 demonstrations (4.11), both code-agent pairs (3.58) and 9 of the 15 ALFWorld pairs. Pairs whose tasks share a sequence of subgoals stay below $R = 3$, such as heating against cooling an object in ALFWorld (1.48, the smallest ratio of all pairs) and put next to against a pick-up followed by a go-to in BabyAI (2.92). In BridgeData 9 of the 21 pairs are separated: the demonstrations of one skill vary widely, and the large baselines (1.23–1.99, against at most 1.39 for ALFWorld and 0.61 for ETO) lower R. Without the surface coordinates (panel b), 34 pairs are separated. The estimate $\widehat{W}_1$ on Φ_D thus separates 33 of the 52 pairs of groups of reference trajectories, among them most of those that differ in environment, instruction kind or demonstrator, and it estimates the distance $W_1(\nu_S, \nu_T)$ of the shift term.

M.2.2 Experiment 3: One Model in Two Agent Harnesses

The code-agent pairs of Experiment 2 change the harness and the model together. Experiment 3 holds the model fixed, changes only the harness, and compares the model's own episodes, which are the trajectories that run-time monitors, verifiers and trajectory filters read. We run Qwen3-8B [Qwen Team, 2025], an open-weight model with eight billion parameters, in bf16 with its thinking mode disabled, served by vLLM [Kwon et al., 2023] in two harnesses with the same sampling settings (temperature 0.7, top-p 0.8, top-k 20, at most 4,096 tokens per response) and at most 100 model calls per episode: mini-SWE-agent 2.4.6 [Yang et al., 2024], which gives the model a single shell tool, and OpenHands, which has separate tools for shell commands and for viewing and editing files. The tasks are 300 of the 500 instances of SWE-bench Verified [Jimenez et al., 2024, OpenAI, 2024], human-checked GitHub issues of Python repositories, drawn in proportion to the number of instances per repository. Each instance and harness gets one episode in a fresh container built from

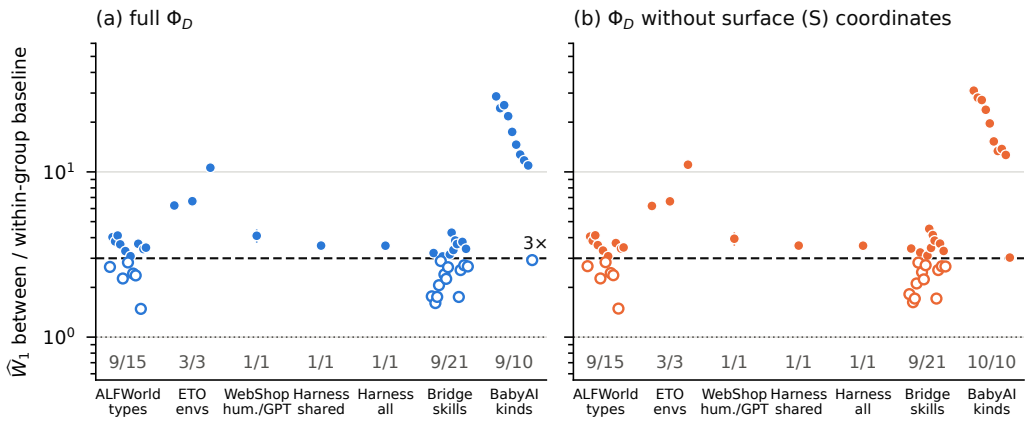


Figure 4: Structural shift between groups of reference trajectories (Experiment 2). Vertical axis: ratio R of the between-group $\widehat{W}_1$ to the larger of the two within-group baselines, on a logarithmic scale. One marker per pair of groups, spread sideways within a column. (a) All coordinates of Φ_D. (b) Without the surface coordinates (S). Columns from left to right: the six ALFWorld task types, the three ETO environments, human against GPT-4 demonstrations in WebShop (hum./GPT), SWE-agent against OpenHands, in which harness and model both differ, on the issues resolved by both (Harness shared) and on all resolved issues (Harness all), the seven Bridge skills and the five BabyAI instruction kinds. Filled marker: separated pair. Hollow marker: pair not separated. Dashed line, labelled $3\times$: $R = 3$. Dotted line: $R = 1$. Bars: standard deviation over seeds of the between-group $\widehat{W}_1$, divided by the larger baseline; most are hidden by the markers. Numbers at the bottom: separated pairs out of all pairs of the column.

the official image of the instance. We project the episodes onto the 16 coordinates of the code-agent projection of Experiment 2 and compute R with ten seeds and $N = 150$, half the episodes of each harness.

Between the harnesses $\widehat{W}_1 = 3.81$, with a coefficient of variation of 0.024, against baselines of 1.48 for mini-SWE-agent and 1.74 for OpenHands, so $R = 2.19$. Computed on each coordinate alone, $\widehat{W}_1$ between the harnesses exceeds that coordinate's own baseline by a factor of 1.95 to 11.5. The largest differences concern reading code, where the harnesses offer different tools. Under mini-SWE-agent the model makes 50.4 read calls per episode against 17.3 under OpenHands and 73.8 tool calls in all against 43.5. The success changes as well. On the 299 instances whose reference fix passes the tests, the resolve rate rises from 1.7% under mini-SWE-agent to 6.4% under OpenHands, by 4.7 points (95% confidence interval $[2.0, 7.7]$, McNemar test $p = 0.001$).

With Qwen3-8B held fixed, the change of harness thus moves the trajectories along every coordinate and changes how often the agent succeeds, so a monitor, verifier or trajectory filter calibrated on one harness receives shifted inputs under the other.

M.3 Lemma 2 under Shifted Position Indices

Lemma 2 states that shifting the position indices of a sub-trajectory changes the attention scores of an APE model almost surely, while under RoPE they stay the same (Appendix C.3). APE adds a learned vector for each position to the embedding of the token at that position (Appendix C.1), RoPE [Su et al., 2024] rotates queries and keys so that a score depends on the positions only through their distance, and ALiBi [Press et al., 2022] subtracts from each score a slope per head times the distance. Experiments 4–6 shift all position indices without inserting a token, so the tokens and their distances stay the same, and in exact arithmetic every attention score under RoPE and every attention weight under ALiBi is unchanged.

Experiments 4 and 5 use 33 public base models, trained only to predict the next token. Twelve use APE: GPT-2 [Radford et al., 2019] with 124M to 1.5B parameters, GPT-Neo [Black et al., 2021] with 125M to 2.7B and OPT [Zhang et al., 2022] with 125M to 6.7B. Five use ALiBi: BLOOM

[BigScience Workshop, 2022] with 560M to 7.1B. Sixteen use RoPE: six Pythia models [Biderman et al., 2023] with 160M to 6.9B, GPT-J 6B [Wang and Komatsuzaki, 2021], OLMo-2 [Team OLMo et al., 2024] with 1B and 7B, SmolLM2 [Ben Allal et al., 2025] with 135M to 1.7B, and Qwen2.5 [Qwen Team, 2024] with 0.5B to 7B.

M.3.1 Experiment 4: Attention Scores of Pretrained Models

Experiment 4 compares the attention scores of full pretrained architectures with and without a shift of all position indices, to see whether those of APE models change substantially and those of RoPE models stay unchanged up to rounding. Eight of the 33 models take part: GPT-2 124M and 355M and GPT-Neo 125M and 1.3B with APE, and Pythia 160M, 410M and 1.4B and SmolLM2 360M with RoPE. Each model reads 512 sequences of 512 tokens, 128 from each of four families: Countdown reference trajectories of μ_{DFS} and of μ_{BFS}, Dyck-4 strings of four kinds of brackets nested at most 12 deep, and source files of the Python 3.10 standard library [Python Software Foundation, 2025]. Each sequence runs once with position indices $0, \ldots, 511$ and once with $k, \ldots, 511 + k$ for $k \in \{16, 64, 256\}$, within the context of every model (96 combinations of model, shift and family). All runs are in fp32 with the TF32 mode of the GPU turned off, and each is repeated in float64 to separate real changes from rounding. We record the largest change of the attention scores before the softmax, divided by the largest absolute score, in the first layer and over all layers, the share of positions whose top-1 next-token prediction changes, and the change of the log-likelihood of the text in nats per token.

In all 48 combinations with an APE model, the largest first-layer change is 0.42–1.19 of the largest absolute score, the same order as the scores themselves. The shift changes 14.0–87.4% of the top-1 predictions and lowers the log-likelihood by 0.20–6.77 nats per token. Under RoPE the change in fp32 is at most 6.2×10^{-6} in the first layer and 3.2×10^{-3} over all layers, about the difference between the fp32 and float64 computations without any shift, and in float64 it is at most 4.4×10^{-12}, so the fp32 change is rounding. RoPE changes at most 0.038% of the top-1 predictions, all at nearties between the two most likely tokens. The contrast of Lemma 2 thus holds in the attention scores and predictions of these eight pretrained models.

M.3.2 Experiment 5: Answers under Shifted Position Indices

Experiment 5 applies the shifts of position indices that occur when a model is served (through padding, cached prefixes and sliding windows) and checks whether they change the answers of APE models and leave those of RoPE and ALiBi models unchanged. Each of the 33 models answers $3{,}000$ items, scored by exact match: 500 for each of five few-shot tasks (copying 6–24 words, reversing 2–4 words, closing a Dyck-4 prefix, adding two numbers of 1–2 digits, and completing a line of a Countdown search trace) and 500 key–value retrieval items whose prompts grow with the context of the model, up to $3{,}000$ tokens. Decoding is greedy with cached keys and values, in fp32 except in condition (vi) below, and each item is decoded in the same batch as its unshifted version.

We shift the position indices by $k \in \{16, 64, 256, 1024\}$, within the trained context, in four ways that occur in serving: (i) all indices; (ii) left padding with k tokens and position indices that count the padding; (iii) a cached prefix, the prompt up to its final question, computed k positions later; and (iv) a sliding window with attention sinks [Xiao et al., 2024], which keeps the 128 most recent cache slots and the first four, shifted by k and compared with the same window without the shift (few-shot tasks only). In the control (v), a cached prefix of P tokens keeps its indices and the rest of the prompt starts at $P + k$, so every distance between the prefix and a later token grows by k. Condition (vi) repeats (i) in bf16. A change in accuracy is significant when its paired bootstrap 95% confidence interval over items excludes 0.

Figure 5 shows the results. In fp32 under (i)–(iv), the 21 ALiBi and RoPE models change 40 of their $948{,}000$ answers, and every accuracy changes by at most 0.0003. The twelve APE models lose 0.13–0.52 in accuracy at $k = 256$ under (i)–(iii) and 0.04–0.50 under (iv), and under (i) at $k = 256$ they change 70–100% of their answers. In the control (v), the answers change under all three encodings, by up to 0.44 in accuracy at $k = 64$, and the change is significant for 63 of the 64 combinations of RoPE model and shift and for 19 of the 20 of ALiBi model and shift. In bf16 the APE losses remain. The RoPE models other than Pythia 160M (not shown in Figure 5c) stay within 0.022 of their unshifted bf16 accuracy, with significant changes only for Qwen2.5 1.5B and 7B at $k \geq 64$. The ALiBi models lose 0.007–0.033 at $k = 256$ and 0.28–0.36 at $k = 1024$,

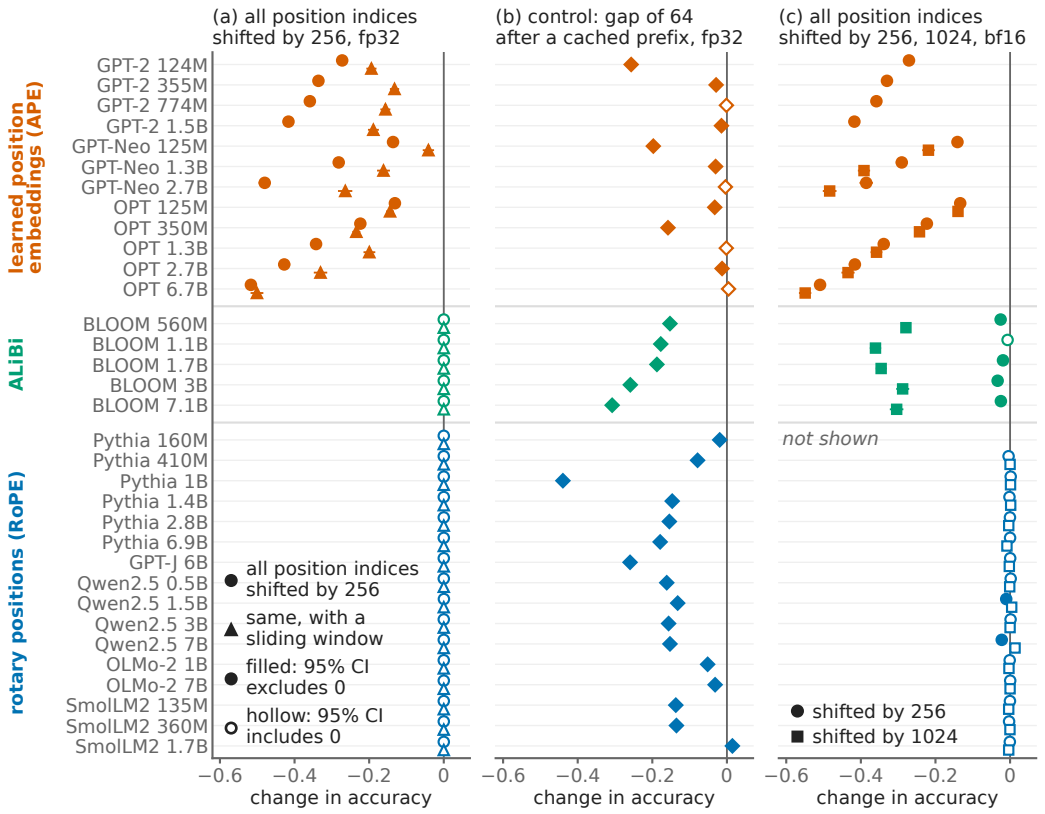


Figure 5: Change in accuracy caused by shifting the position indices, for the 33 models of Experiment 5, one row per model, grouped and colored by positional encoding (APE, ALiBi, RoPE). Horizontal axis: accuracy with the shift minus accuracy without it. Bars: paired bootstrap 95% confidence intervals over items, mostly hidden by the markers. Filled marker: the interval excludes 0. Hollow marker: it includes 0. (a) fp32. Circles: all position indices shifted by 256, condition (i). Left padding (ii) and a cached prefix placed 256 positions later (iii) give the same changes for the APE models and changes of at most 0.0003 for the others, and are not drawn. Triangles: the same shift inside a sliding window (iv), compared with the sliding window without the shift (few-shot tasks only). (b) Control (v), fp32, diamonds: the cached prefix keeps its position indices and the rest of the prompt starts 64 positions later. (c) bf16: all position indices shifted by 256 (circles) and by 1024 (squares, where the shifted prompt fits the trained context). Pythia 160M is not shown in (c), since its bf16 accuracy without the shift is 0.012.

consistent with BLOOM adding the slope times the absolute position of the key, which equals the slope times the distance up to one constant per query but has a bf16 rounding error that grows with the position index. The shifts of position indices that occur in serving thus change the answers of the APE models, while the ALiBi and RoPE models in fp32 keep theirs as long as the distances between tokens are kept.

M.3.3 Experiment 6: Two Countdown Models of Section 5

Experiment 6 turns to the two models of Appendix J, f_1 (RoPE, $L = 12$) and f_2 (APE, $L = 48$): we shift all their position indices in fp32 and record the effect on their routing decisions and their success on Countdown. The routing decisions are the node identifier after `Moving to Node` and the operation after `Exploring Operation` (Appendix K.4). We use those after the first leaf-level operation line of each trajectory, a line that explores an operation on a node with two numbers, such as `Exploring Operation: 3+10=13, Resulting Numbers: [13]`. After this line the model must decide where the search goes next. The line comes before the end of the search and directly after a dead end at the same node or a return to that node, and the node still has an unexplored opera-

tion whose result misses the target. In the teacher-forced test, the model reads a reference trajectory, and we compare its routing predictions after this line with and without a shift of all position indices by 16 or 64. In the free-running test, the model generates its own trajectory, generation resumes at this line with all indices shifted by 16 or 64, and the outcome is scored as in Section 5.

On the 2,863 reference trajectories of the three evaluation sets in which such a line, followed by at least one routing decision, lies within the first 896 tokens, the shift changes none of the 94,278 routing predictions of the RoPE model, and 25.7% (shift 16) and 28.3% (shift 64) of those of the APE model. The free-running test uses the first 1,000 problems of each evaluation set and applies to the 2,128 problems for the RoPE model and the 1,143 for the APE model whose output contains such a line with at least 64 of the 512 tokens of the generation budget left. For the RoPE model the outcome of none of these problems changes (success 0.594). For the APE model, success falls from 0.527 by 0.40 (shift 16) and by 0.39 (shift 64). The contrast of the lemma thus carries over to the task success of two models of the paper, consistent with the mechanism of Assumption 2, in which changed routing decisions turn into failed searches.

M.4 Splitting Deeply Nested Tasks into Separate Model Calls

Lemma 3 maps a Countdown search to a word of brackets, and any task with nested subgoals can be written in the same way. Lemma 5 bounds the risk from below when the nesting that one forward pass must resolve exceeds $\alpha_{\mathrm{circ}}L$, which gives the depth term. Agent systems often spread deep nesting over several calls, for example by passing a subtask to a sub-agent. Experiments 7 and 8 measure whether such splitting restores success and how deep each call should be.

In such a word, `( a` opens a subgoal of type `a` and `) a` closes it, and at every closing bracket the model predicts the type that is closed. With calls of depth at most 2, the plan `( a ( b ( c ( d ) d ) c ) b ) a` becomes the call `( a ( b ) b ) a`, in which `b` is an empty subgoal, followed by `( b ( c ) c ) b` and `( c ( d ) d ) c`, each in a fresh context. A task is solved only if every prediction of every call is right, and the program that splits it uses only its nesting structure. On a calibration set, separate from the test set, we measure how often one call solves a task of each depth, fit to these rates a curve that does not increase with depth, and read from it d_{ext}, the largest depth at which one call succeeds at least half the time, and $\hat{d}$, the largest depth at which it succeeds at least 80% of the time. Figure 6 shows the results.

M.4.1 Experiment 7: Nested Plans

Experiment 7 trains small models on nested plans to find out whether splitting restores success beyond d_{ext}, and whether d_{ext} follows the number of layers, as the threshold $\alpha_{\mathrm{circ}}L$ suggests, or the depth of the training plans. A plan has 48 subgoals of 8 types and maximum nesting depth exactly d: a spine of d nested subgoals and $48 - d$ further subgoals, each attached at random to a subgoal of depth below d or to the top level. Every plan has 194 tokens whatever its depth, and it is solved when the type predicted at each of its 48 closing brackets is correct, all other tokens being given.

We train the GPT-NeoX (RoPE) and GPT-Neo (APE) architectures of Appendix K in five shapes, width 256 with $L \in \{2, 4, 8, 12\}$ layers and a two-layer model of width 624 whose number of parameters matches that of the 12-layer model within 1.3%, on plans whose depth is drawn uniformly from 1 to $d_{\mathrm{train}} \in \{2, 4, 6, 8\}$. Training takes 30,000 steps of 256 plans (AdamW, peak learning rate 10^{-3} with cosine decay, loss on the predicted types), and the 40 configurations (two encodings, five shapes, four training depths) with three seeds each give 120 runs. Five configurations in which some run stays below a single-call success of 0.9 at a depth up to d_{train} are trained again for 60,000 steps, after which three of them reach 0.9 at every such depth in every run. For each run, d_{ext} comes from 200 calibration plans per depth. A plan deeper than d_{ext} is split into calls of depth $D = \min(d_{\mathrm{ext}}, d_{\mathrm{train}})$, which keeps every call within the training depths and needs $D \geq 2$. The test sets have 500 plans per depth, up to depth $4d_{\mathrm{train}} + 8$.

In 38 of the 40 configurations, every run reaches a single-call success of at least 0.9 at every depth up to d_{train}. In all 114 runs of these configurations, splitting gives a mean success of at least 0.987 over the depths beyond d_{ext}, and at least 0.91 at each of them, against at most 0.04 for one call, as Figure 6b shows for the 12-layer RoPE model. d_{ext} follows d_{train}: it is 2, 4–5, 6–9 and 9–12 for $d_{\mathrm{train}} = 2$, 4, 6 and 8, with no systematic dependence on the number of layers, the width or the positional encoding, and exceeds d_{train} by at most four levels (Figure 6a). Deeper training saves

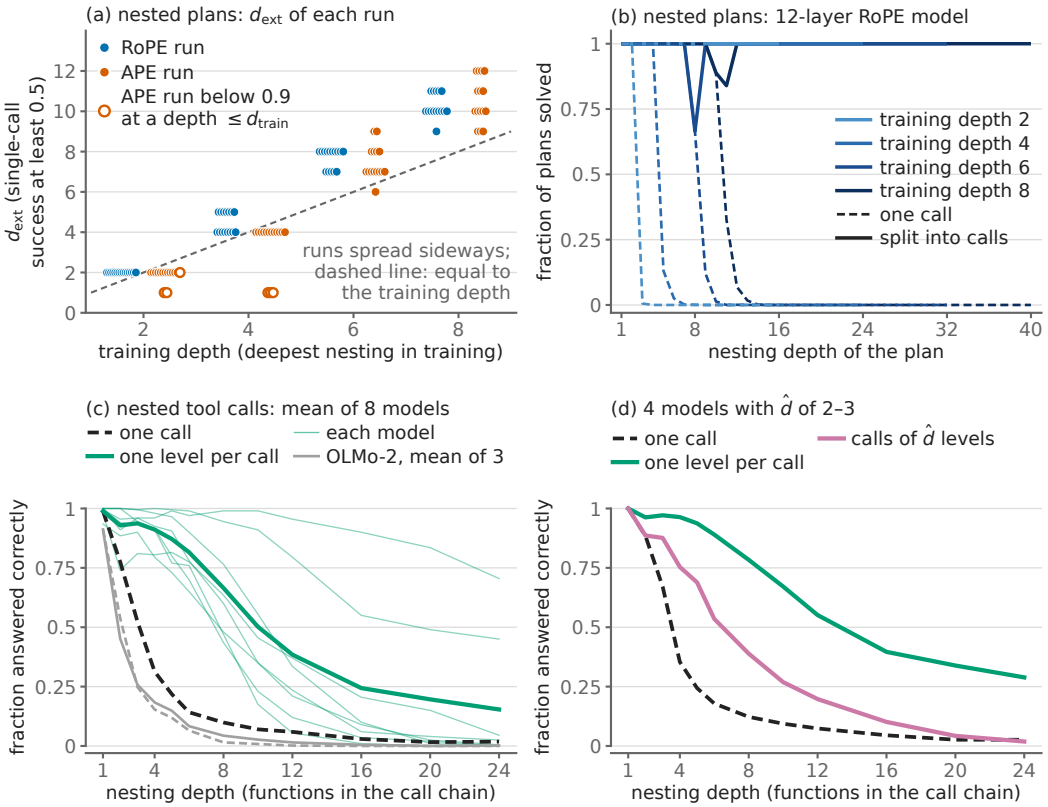


Figure 6: Splitting deeply nested tasks into separate model calls (Experiments 7 and 8). (a) Experiment 7: d_{ext}, the largest depth that one call solves at least half the time, for each of the 120 runs (vertical axis), against the training depth d_{train} (horizontal axis). RoPE runs are drawn left and APE runs right of each training depth, and runs with equal values are spread sideways. Dashed line: $d_{\text{ext}} = d_{\text{train}}$. Hollow markers: the six runs of the two-layer width-624 APE model at $d_{\text{train}} = 2$ and 4, whose single-call success stays below 0.9 at some depth up to d_{train}. (b) Experiment 7, 12-layer RoPE model: fraction of plans solved (mean of 3 seeds) against the nesting depth of the plan, one shade per training depth. Dashed: one call. Solid: plans deeper than d_{ext} split into calls of depth $\min(d_{\text{ext}}, d_{\text{train}})$, and other plans solved in one call. The dips between d_{train} and d_{ext} come from plans that are still solved in one call. (c, d) Experiment 8: fraction of the 200 test chains answered correctly against the depth d, the number of functions in the chain. (c) Mean of the eight Qwen, Phi and Mistral models with one call (dashed) and with one level per call (thick solid green). Thin lines: each of these models with one level per call. Gray, with the same line styles: mean of the three OLMo-2 models. (d) Mean of the four models whose $\hat{d}$ is 2 or 3: one call (dashed), one level per call (green) and calls of $\hat{d}$ levels (pink).

calls: beyond d_{ext}, a plan needs 20.8 calls on average at $d_{\text{train}} = 2$ and 4.5–4.6 at $d_{\text{train}} = 8$. The two other configurations are the two-layer width-624 APE model at $d_{\text{train}} = 2$ and 4: five of their six runs have $d_{\text{ext}} = 1$, too shallow for a split with $D \geq 2$, and in the sixth splitting raises the success beyond $d_{\text{ext}} = 2$ from 0.00 to 0.998. In models of 2 to 12 layers, the depth that one call resolves thus follows the training depth whatever L is, and splitting into calls within the training depths restores success on deeper plans.

M.4.2 Experiment 8: Nested Tool Calls with Open-Weight Language Models

Experiment 8 measures $\hat{d}$ for pretrained language models, whose training depth is unknown, on a task shaped like the tool calls of an agent: a registry lists one-line Python functions, and a question asks for the value of a chain of nested calls, such as `bell(4)` for the registry

```
def lamp(x): return x + 2
def door(x): return lamp(x) + 3
def bell(x): return door(x) + 1
```

a chain of depth $d = 3$ whose value is 10. With one level per call, the chain is split from the innermost function outward: the first call returns `lamp(4)` $= 6$, and the next call sees the line of `lamp` replaced by this value and asks for `door(4)`. Only the answer of the last call is scored.

A registry has 26 lines: the d functions of the chain, named by common English nouns and each adding a constant $c \in \{1, 2, 3\}$, two distractor functions, and $24 - d$ comment lines that keep the length of the prompt the same for every d. Four worked examples precede every prompt. The model may generate at most eight tokens, and the first integer among them is scored by exact match, so the model must follow the chain within its forward passes. The calibration set has 100 and the test set 200 items for each $d \in \{1, 2, 3, 4, 5, 6, 8, 10, 12, 16, 20, 24\}$. Eleven instruction-tuned models answer, in bf16 with greedy decoding: Qwen2.5 [Qwen Team, 2024] (7B, 14B and 32B parameters), OLMo-2 [Team OLMo et al., 2024] (7B, 13B, 32B), Qwen3 [Qwen Team, 2025] (8B, 14B, 32B, thinking mode disabled), Phi-4 14B [Abdin et al., 2024] and Mistral-Small-24B-Instruct-2501 [Mistral AI, 2025]. We compare four ways of answering: one call; one call whose prompt also lists the open call chain, such as `bell(4) -> door(4) -> lamp(4)`; one level per call, with d calls; and calls of $\hat{d}$ levels, with $\lceil d/\hat{d} \rceil$ calls and $\hat{d}$ measured for each model on the calibration set.

The accuracy of one call falls quickly as the chain grows (Figure 6c), from 0.88–1.00 at $d = 1$ to at most 0.52 at $d = 4$ and at most 0.15 at $d = 8$. $\hat{d}$ is 3 for Qwen2.5 32B, 2 for Qwen3 14B, Phi-4 and Mistral-Small-24B, and 1 for the other seven models. Listing the chain changes the single-call accuracy by at most 0.09 on average over depths. One level per call raises the accuracy of the eight Qwen, Phi and Mistral models at $d = 8$ from 0.01–0.15 to 0.43–0.99, and Mistral-Small-24B answers 0.70 of the chains correctly at $d = 24$. For the three OLMo-2 models, one level per call gives 0.04–0.06 at $d = 8$, against 0.00–0.03 in one call, and 0.37–0.59 at $d = 2$, against 0.43–0.72. For the four models whose $\hat{d}$ is 2 or 3 (Figure 6d), calls of $\hat{d}$ levels need 4 or 3 calls at $d = 8$ and reach 0.24–0.51, against 0.43–0.99 with one level per call. A chain is right only if every call is right, so the errors of the calls compound, and by the definition of $\hat{d}$ a call of $\hat{d}$ levels errs in up to a fifth of the cases. One level per call thus beats calls of $\hat{d}$ levels at $d = 8$ for all four models where the two differ, since each of its calls stays at a depth where one call is almost always right.

M.5 Summary of the Eight Experiments

Table 3 summarizes the eight experiments. Each concerns one part of Theorem 2: Experiment 1 the balance between the residual ϵ and the estimation of W_1, Experiments 2 and 3 the distance W_1 in the shift term, Experiments 4–6 the factor $K_{f,\mathrm{PE}}$ through Lemma 2, and Experiments 7 and 8 the depth term.

Table 3: Results of the eight experiments of Appendix M. *Experiment*: number, short name and section. *Setting*: the models and data. *Finding*: the main numbers and what they show.

Experiment	**Setting**	**Finding**
1. More coordinates on Countdown (M.1.1)	12, 16, 24 and 34 coordinates; f_1 and f_2; training trajectories of the three measures	ROC-AUC changes by at most 0.003, while $\widehat{W}_1$ between measures grows by a factor of 1.8–2.0; the twelve coordinates are the better choice
2. Groups of agent, embodied and robot trajectories (M.2.1)	52 pairs of groups from the ETO expert trajectories, the Nebius code-agent runs, BridgeData V2 and BabyAI	33 pairs separated (34 without surface coordinates); tasks that share a sequence of subgoals give the smallest ratios
3. One model in two agent harnesses (M.2.2)	Qwen3-8B in mini-SWE-agent and OpenHands, 300 SWE-bench Verified instances	$R = 2.19$, and every coordinate moves by 1.95–11.5 times its own baseline; the resolve rate rises from 1.7% to 6.4%
4. Attention scores of pretrained models (M.3.1)	Four APE and four RoPE models; shifts of 16, 64 and 256	Under APE the first-layer change is 0.42–1.19 of the largest score; under RoPE the change is rounding (at most 4.4×10^{-12} in float64)
5. Answers under shifted position indices (M.3.2)	33 public models; all indices shifted, left padding, cached prefix, sliding window	In fp32 the ALiBi and RoPE models change 40 of $948{,}000$ answers, and the APE models lose 0.13–0.52 in accuracy at $k = 256$; a gap after a cached prefix changes the answers under every encoding
6. Two Countdown models (M.3.3)	f_1 (RoPE) and f_2 (APE) of Section 5	No outcome of the RoPE model changes; the success of the APE model falls by 0.40 and 0.39 (shifts 16 and 64)
7. Nested plans (M.4.1)	120 runs (40 configurations, three seeds) trained from scratch, $d_{\mathrm{train}} \in \{2, 4, 6, 8\}$	In the 114 runs that learned their training depths, splitting gives a mean success of at least 0.987 beyond d_{ext}, against at most 0.04 for one call; d_{ext} follows d_{train}
8. Nested tool calls (M.4.2)	11 open-weight models of 7B–32B parameters	With one level per call, the accuracy of eight models at $d = 8$ rises from 0.01–0.15 to 0.43–0.99; calls of $\hat{d}$ levels reach 0.24–0.51

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